



BADAN PUSAT STATISTIK



SENSUS
EKONOMI
2026



UTILIZING MOBILE POSITIONING DATA FOR OFFICIAL STATISTICS

57th UNSC Side Event: Global Push to Put Mobile Phone Data to Work for Statistics

New York, March 4TH 2026

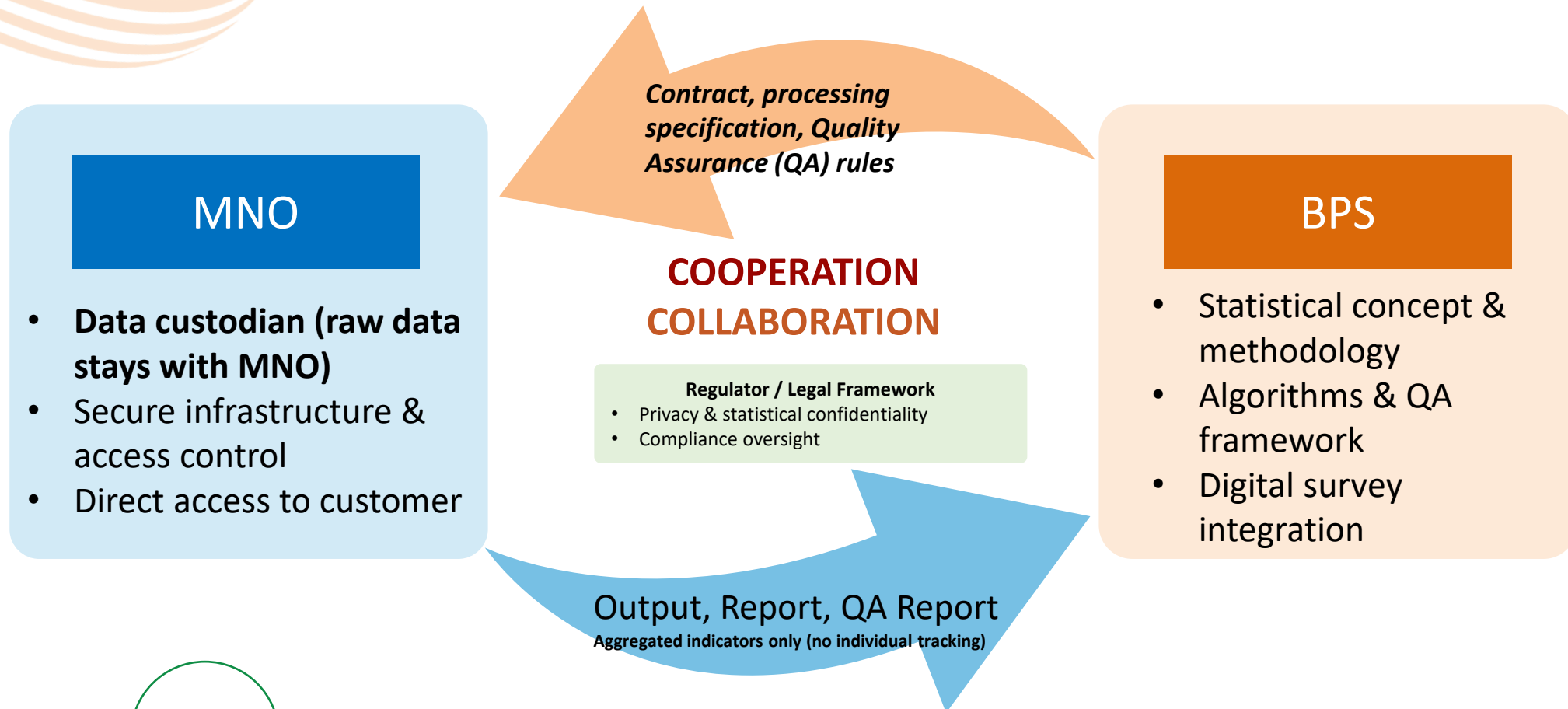


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PARTNERSHIP MODEL IN MPD UTILIZATION



Mobile Network Operators in Indonesia

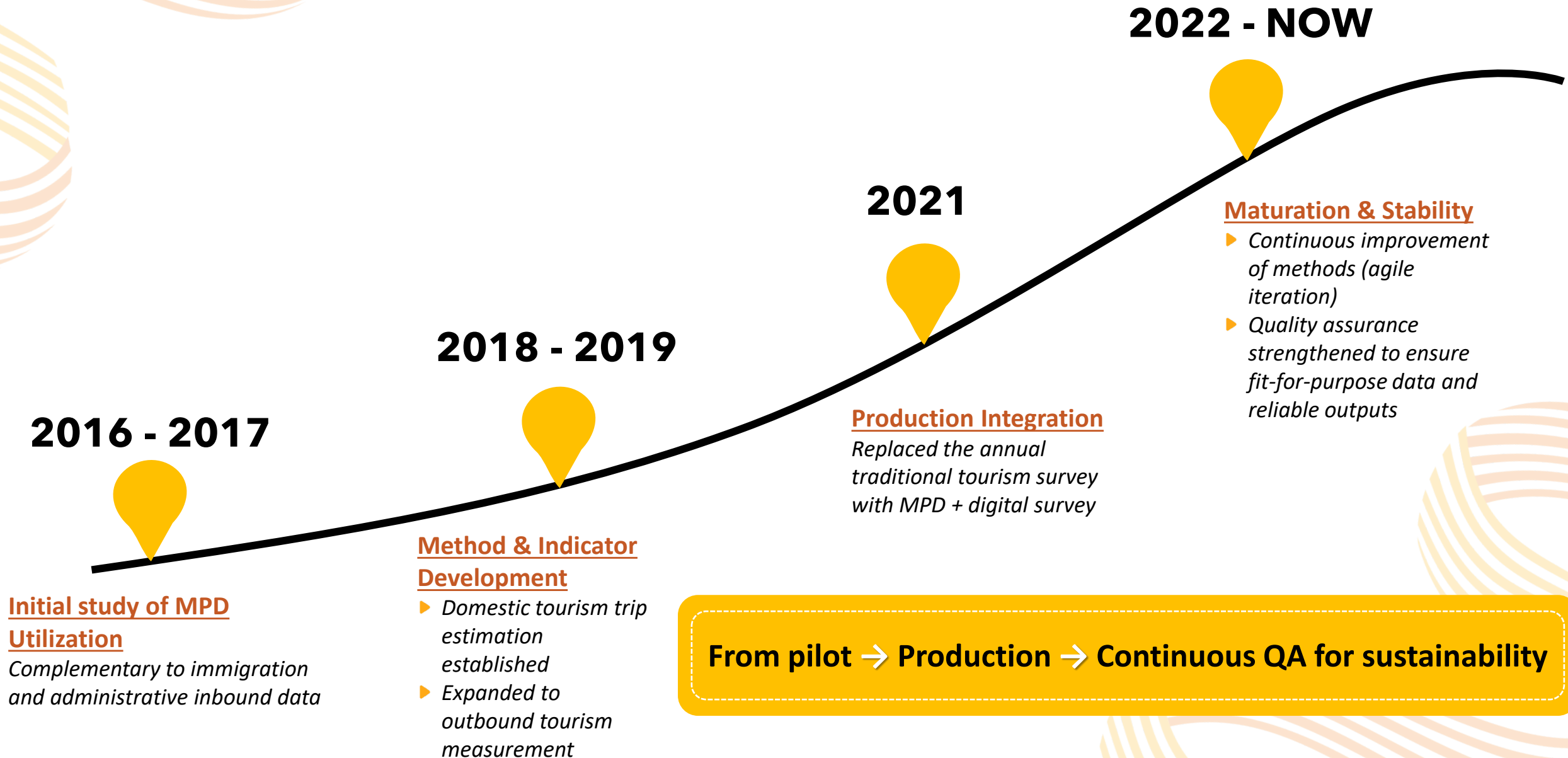


Telkomsel



XLSMART

MPD JOURNEY & SUSTAINABLE PIPELINE



MPD IMPLEMENTATION FOR OFFICIAL STATISTICS

MPD provides aggregated mobility patterns derived from mobile network and device-generated logs, processed under strict statistical confidentiality.

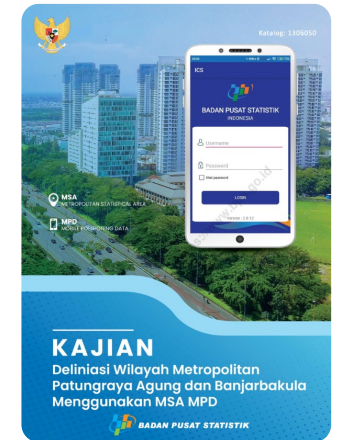
This enables BPS to produce timely and granular indicators for official statistics.



BPS has compiled several statistics produced from utilizing MPD, such as:

- ✓ **Tourism: domestic, outbound, inbound**
- ✓ **Mobility: MSA, commuter, migration, dynamic population**
- ✓ **Special analyses: peak-season/event mobility (Eid, year-end, etc.), mobility during disaster**

1 Annual Publications



2 Monthly Official Statistics News (BRS): Tourism Statistics



- ▶ Inbound Tourism
- ▶ Outbound Tourism
- ▶ Domestic Tourism





BEYOND TOURISM: WHAT MPD CAN UNLOCK

Two signal families (Mobility and ICT) translated into official statistics.

SUBSCRIBER MOBILITY

Movement patterns emerge from repeated location changes over time—aggregated into flows, stays, and shifts.

- ➔ **Commuting & metropolitan mobility** (flows between home–work areas) 
- ➔ **Migration & residential change** (longer-term relocation signals) 
- ➔ **Population dynamic** (how many people stay in an area over a period)
- ➔ **Event-based mobility** (Eid, year-end, disasters) 

ICT (DEVICE INTERACTION)

Digital interaction leaves usage and network-performance signals—aggregated into access, quality, and coverage indicators.

- ➔ **Internet users/adoption** (active data usage patterns) 
- ➔ **Quality of service** (speed/latency proxies, stability indicators—aggregated) 
- ➔ **Service coverage & gaps** (spatial footprint of connectivity) 



Pilots completed; scaling underway.

INDONESIA CAN SHARE EXPERIENCES FOR MPD THROUGH UN REGIONAL HUB

What we offer (Replicable model)

- Governance framework (privacy-by-design).
- Methodology & Quality Assurance playbook.
- Production-ready workflows.

What we have done (Regional contribution)

- Disseminating best practices.
- Regional hub role/peer learning (Malaysia, Philippines, Cambodia, Timor-Leste, Kazakhstan, Georgia, Brunei Darussalam, Mongolia, Pakistan, Nepal).
- Practical templates and lessons learned.

How we sustain (Capacity building)

- Build a solid multi-disciplinary team.
- Continuous training & knowledge transfer.
- Collaboration with MNOs + stakeholders.

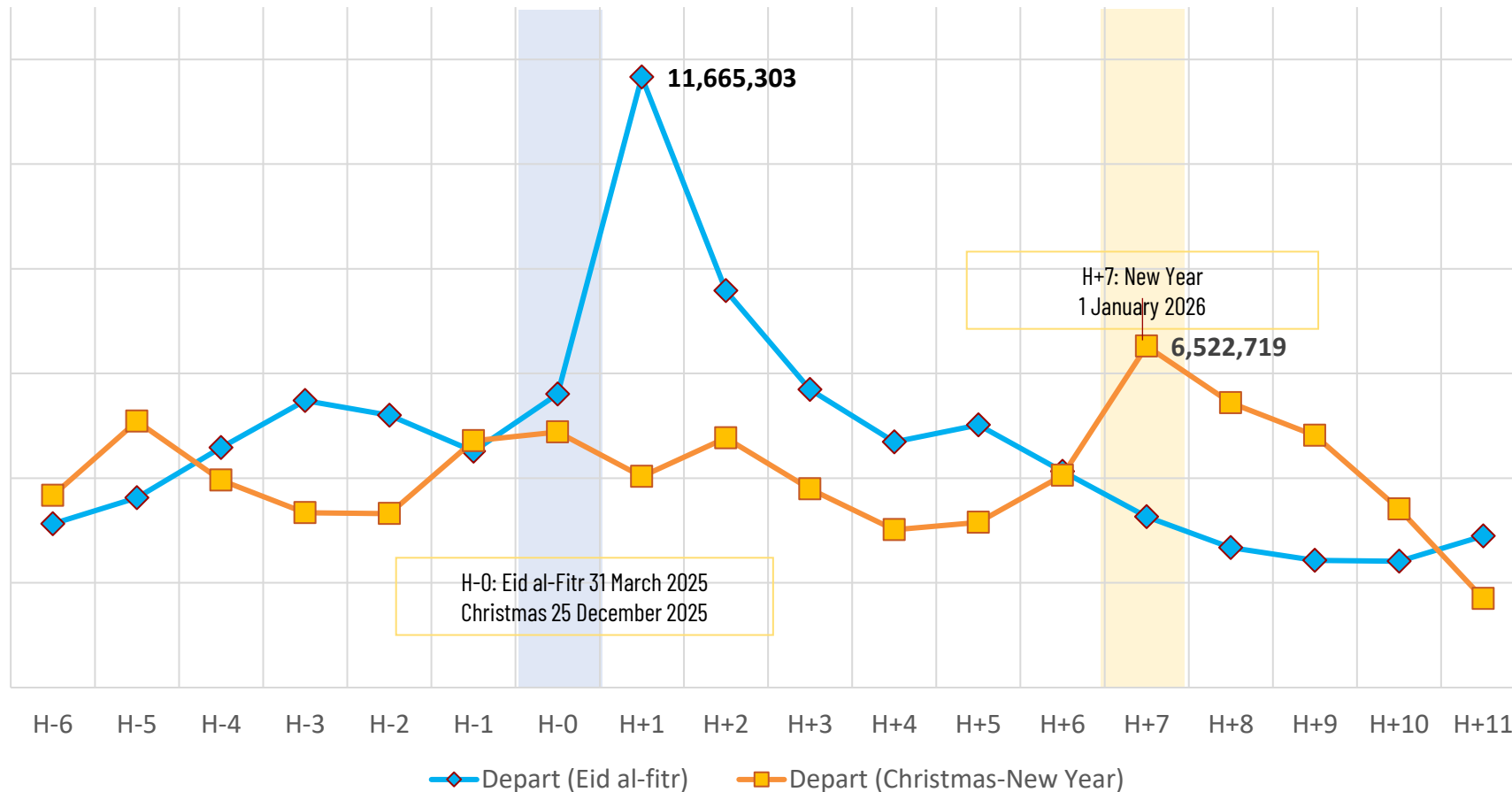
Indonesia is ready to be a learning partner for other countries.

SPECIAL CASE: EID-AL FITR AND CHRISTMAS-NEW YEAR HOLIDAY

Homecoming Flow During Eid 2025 and Christmas-New Year 2025-2026



Number of Domestic Trips During Eid al-Fitr 2025 and Christmas-New Year 2025-2026 (trips)



Source: Mobile Positioning Data (MPD)

Eid-al Fitr and Christmas-New Year

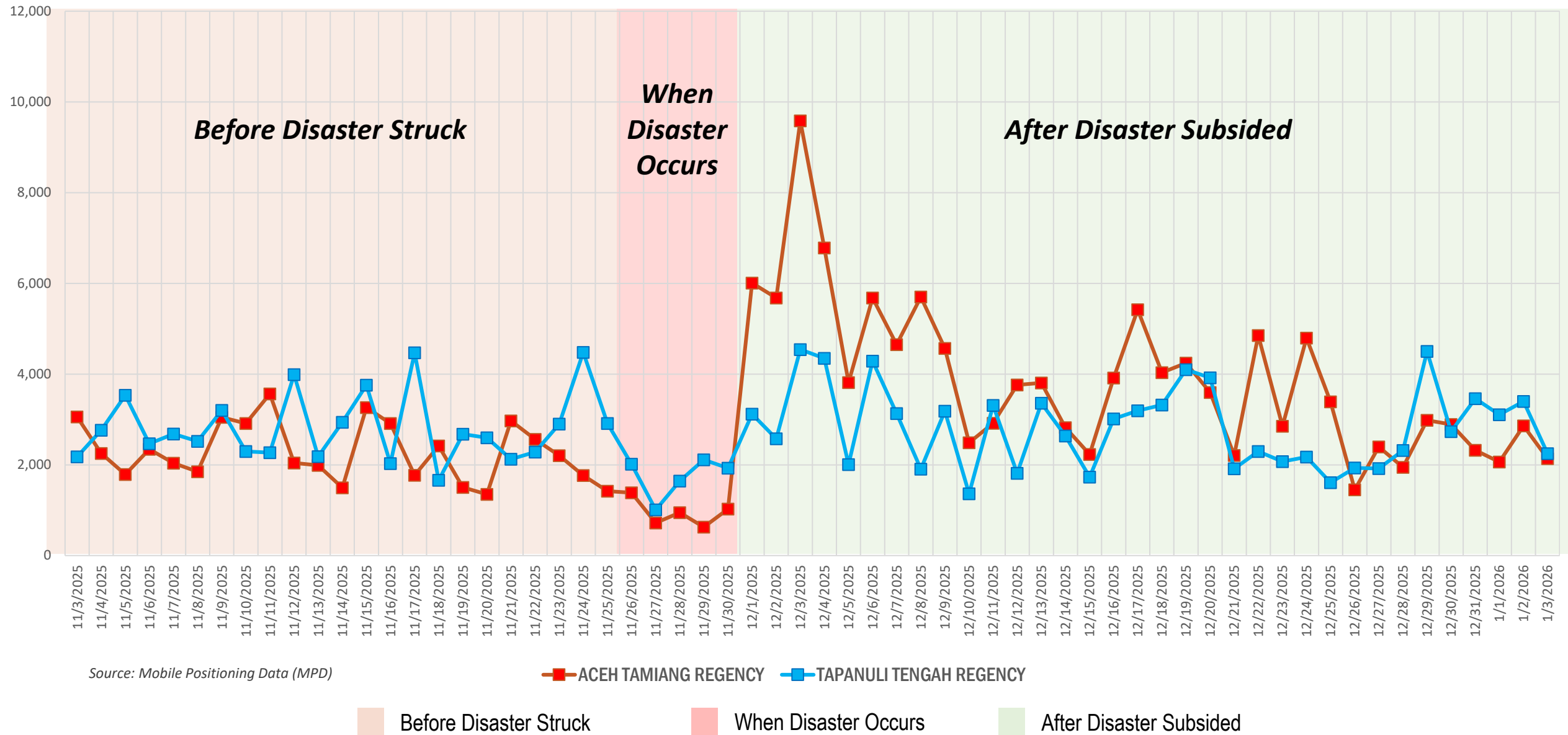
The number of domestic trips during the **Eid al-Fitr 2025 Holiday period (H-6 to H+11)** reached **84,602,924** trips, while trips during the **Christmas and New Year 2025-2026 Holiday period** reached **73,684,329** trips.

The peak of the **Eid al-Fitr 2025** flow was recorded on D+1 with **11,665,303** trips, while the peak of the **Christmas and New Year 2025-2026** flow occurred on H+7 with **6,522,719** trips.



SPECIAL CASE: DISASTER ANALYSIS

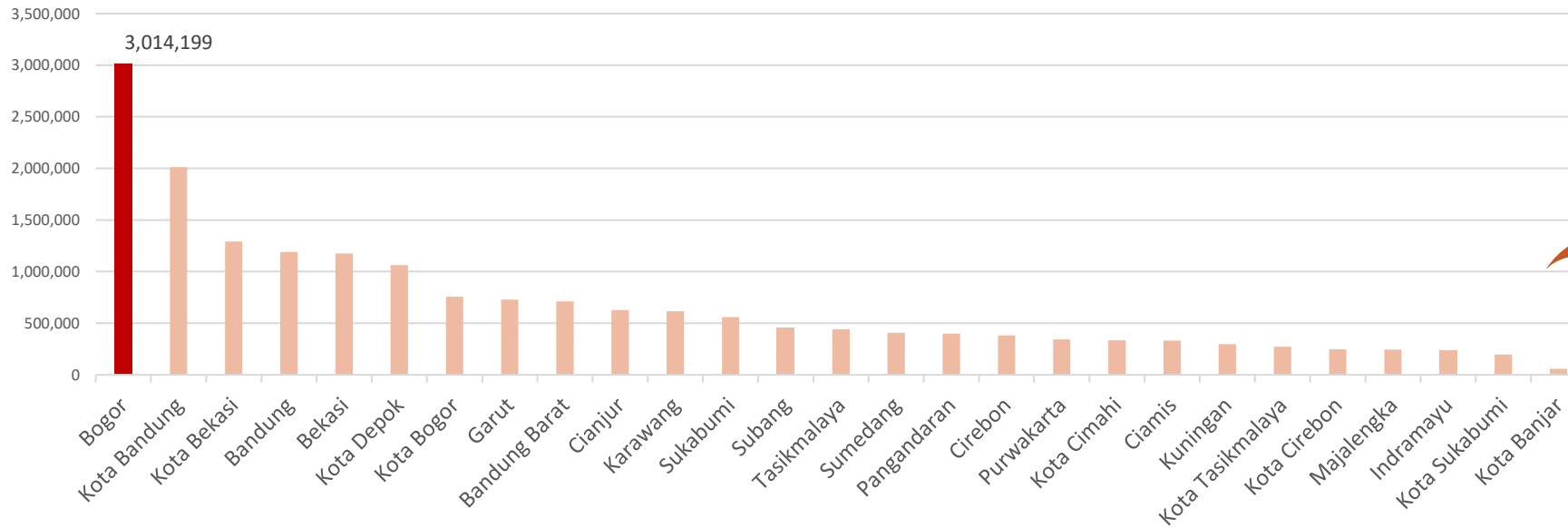
Daily People Mobility **FROM** Disaster-Affected Areas: Aceh Tamiang and Tapanuli Tengah Regency, 3 Nov 2025 – 3 Jan 2026



Source: Mobile Positioning Data (MPD)

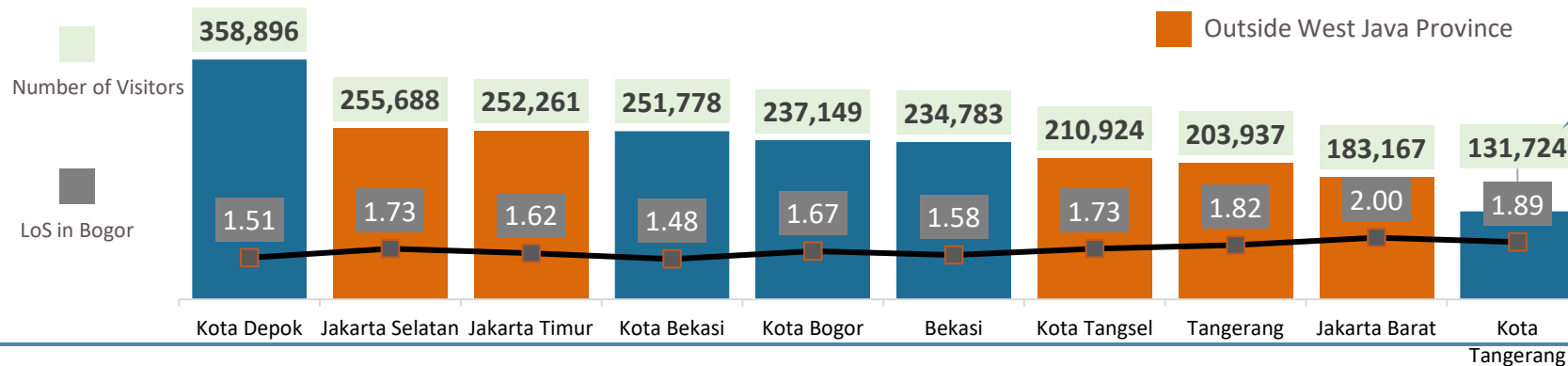
SPECIAL CASE: DOMESTICS TOURISTS TO BOGOR REGENCY, JANUARY 2026

Number of trips by destination in the Province of West Java, January 2026



Number of domestic trip to Bogor reached **3.014.199** trips. It is the **highest among all regencies in the Province of West Java.**

Top 10 Origin Regency/City with Largest Number of Visitors to Bogor and Length of Stay, Jan 2026



Top 3 Province of Origin: **West Java, Jakarta, and Banten**

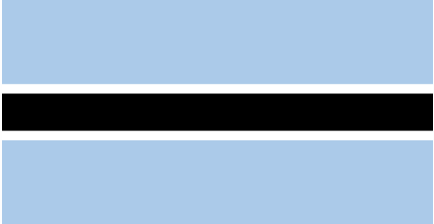
Thank You

**SENSUS EKONOMI 2026
MILIK INDONESIA!**

“Mari Kawal Bersama, Langkah Besar Wujudkan
Kemandirian Perekonomian Bangsa Untuk
Indonesia yang Lebih Sejahtera”

#MencatatEkonomiIndonesia





Botswana's Journey: Leveraging Mobile Phone Data for Official Statistics

Institutional Frameworks, Pipeline Design & Policy Applications

Presented by: Potlako C. Kgari – MPD Project Coordinator

Global Push to Put Mobile Phone Data to Work for Statistics

 4 March 2026 |  New York |  UN Statistical Commission Side Event



STATISTICS BOTSWANA



~2.35 Million

Total Population (Sparsely Populated)



~4.2 Million

Mobile Connections (166%+ Penetration)



581,730 km²

Vast Land Area requiring Coverage

Bridging the Data Gap



Traditional Data Source Limitations

Censuses and household surveys are expensive, geographically limited, infrequent (every 10 years), and slow to process.



The Challenge

Need for real-time, disaggregated, cost-effective statistics to monitor national developmental frameworks.



The Solution

Leveraging continuous digital footprints from mobile networks to generate granular insights.

💡 The Opportunity

High mobile penetration creates a rich, continuous, near real-time data stream. MPD complements and modernizes the National Statistical System (NSS) by providing cost-effective statistics for a vast, low-density environment



1

Awareness & Alignment

2024–2025

Stakeholder Mapping

Internal government consultations mapping the legal landscape: Statistics Act 2009, Data Protection Act 2004, and BOCRA Act 2012 and regulations.

Multi-Stakeholder Workshop

Hosted by SB & GPSDD (March 2024, Gaborone) bringing together key players:

MNOs (Mascom, Orange, BTC)

BOCRA & Data Protection Commission

Ministry of Communications & Dev Partners

Technical partner: Positium

2

Formalizing Arrangements

2025–2026

Memorandum of Understanding

Tripartite MOU between Statistics Botswana, BOCRA, and MNOs establishing:

Purpose of data use

Governance structure

Roles & responsibilities

Dispute resolution mechanisms

Data Sharing Agreements (DSAs)

Tripartite DSA with all MNO specifying:

Data variables to be shared,

Anonymization standards

Delivery frequency & permitted uses

Breach response protocols

3

Pipeline Formalization

2026–2027

Operational Integration

Integrating MPD protocols into Statistics Botswana's operational statistical calendar.

Capacity Building

SADC GDF-MPD Seed Grant application to fund institutional capacity building.

Regional Engagement

Active participation in SADC Regional Workshops (May 2025, Johannesburg).

 **National Stakeholder Convening (Nov 2025)** – Validation of Tourism use case, Formalisation of 2-tier governance structure, (Political and policy consensus, stakeholder alignment)



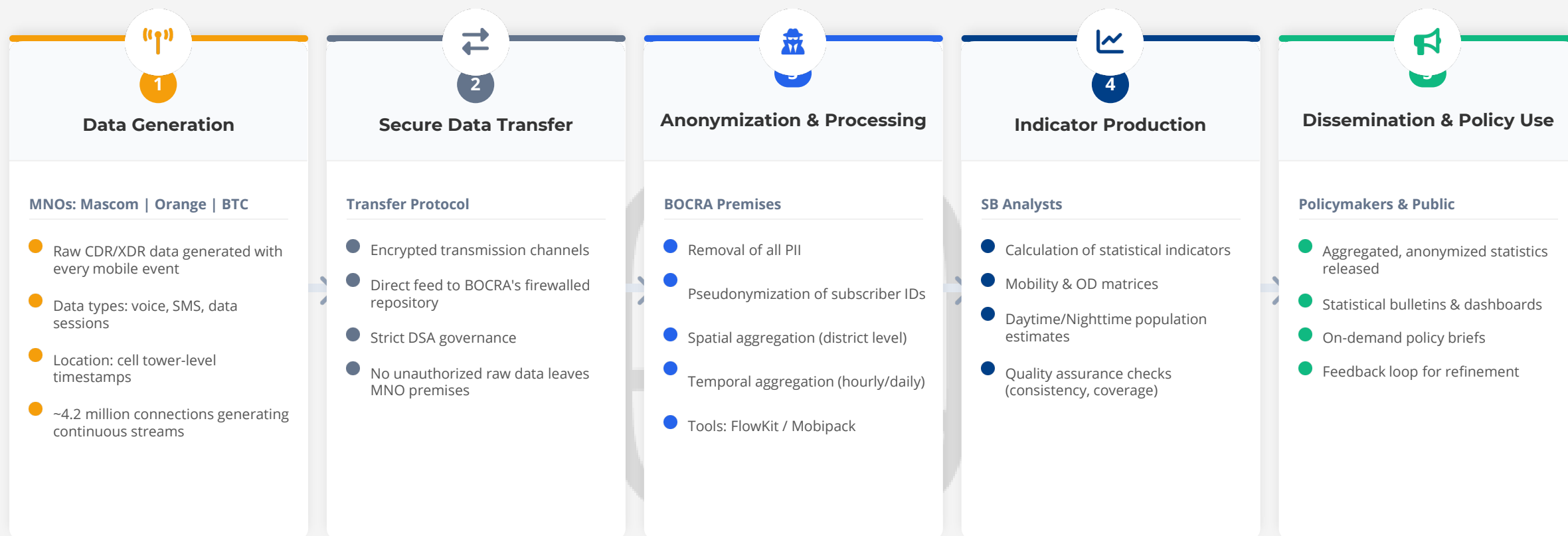
Core Governance Principle

Data stays on regulated premises. Only aggregated, anonymized statistics leave BOCRA's servers.

Botswana's MPD Data Pipeline — From Raw Signal to Policy Insight



STATISTICS BOTSWANA



 **PRIVACY SAFEGUARD:** Individual-level data never leaves the secure pipeline environment. Only aggregated statistics are shared.



Legal Basis

- ✓ **Statistics Act, 2009:** Mandates NSO to collect and use data for official statistics.
- ✓ **Data Protection Act 2024:** Enshrines subscriber rights, data minimization, and purpose limitation.
- ✓ **BOCRA Regulations:** Licensing conditions governing MNO data handling.
- ✓ **SADC Alignment:** Compliance with regional Data Governance Framework.



Privacy Safeguards

- 🔒 **No Individual Access:** Statistics Botswana never accesses raw PII data.
- 🏢 **Secure Premises:** All processing occurs strictly on BOCRA's secure infrastructure.
- 🔗 **Advanced Techniques:** k-anonymity, differential privacy, and spatial aggregation.
- 🔑 **Access Control:** Strict role-based permissions and encrypted audit trails



Governance Oversight

- 👥 **Steering Committee:** Provide strategic leadership and oversight, **Accountability:** transparency and data ethics
- 📅 **Regular Reviews:** Annual assessment of MOU/DSA terms and conditions.
- ✅ **IDPC Approval:** Independent review and approval of all anonymization protocols.
- ⚖️ **Accountability:** Transparent reporting via Statistics Botswana official report.

Trust is paramount: Our framework ensures data utility without compromising citizen privacy.

What Are We Working Toward? Priority Policy Applications



STATISTICS BOTSWANA



Population Dynamics

- Real-time population distribution mapping
- Inter-district migration flows analysis
- Internal displacement tracking
- Intercensal population estimates



Transport & Mobility Planning

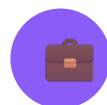
- Origin-Destination (OD) matrices for road planning
- Commuter flow pattern analysis
- Public transport route optimization
- Identification of infrastructure gaps



Tourism Statistics

(Use case alignment with National Dev priorities)

- International + Domestic tourism visitor flows
- Visitor dwell time analysis
- Economic footprint estimation
- Complementing formal tourism surveys



Internal Mobility & Migration

- Rural-Urban migration flow mapping
- Labour mobility & commuting patterns
- Disaster displacement tracking
- Pandemic preparedness & disease outbreak



ICT & Information Society

Estimate ICT sector statistics

- Actual mobile internet usage patterns
- Measuring urban vs. rural digital divide
- Digital inclusion metrics
- Complementing ICT household surveys

 These applications directly support Botswana's National Development Plan (NDP12), Vision 2036, BETP and SDG monitoring.

NDP 12

Vision 2036

SDGs



Challenges



Legal Clarity

Navigating intersections of Statistics Act, Data Protection Act, and BOCRA Act & regulations,.



Trust-Building with MNOs

Data owners needed assurance on confidentiality and liability limits.



Data Security Infrastructure

BOCRA's IT infrastructure require upgrading to host secure processing environment.



Capacity Gaps

Statistics Botswana required training on CDR analytics and big data tools.



Financing the Foundation

Initial investment in infrastructure may require securing external support.



Lessons Learned



Align Stakeholders First

Relationships open doors that laws alone cannot; early engagement is key.



Anchor in Legal Authority

The Statistics Act 2009, provided necessary legitimacy; build on existing mandates.



Keep Data on Premises

The "data stays at BOCRA" principle resolved most MNO privacy concerns.



Regional Learning Accelerates

SADC and other Partner workshops participation help shorten learning curve.



Champion at Political Level

Ministerial / level buy-in was essential to unlock MNO cooperation.

Thank you

Formalizing data access – MoU templates



MPD for Policy & Statistics: Data Access Models

Stakeholders agree on the [architecture of the data access model](#) via the assignment of specific roles.

Role	Stakeholder
Data owner – Who collects and owns MPD?	Always MNOs
Data Custodian – Who hosts the data and maintains infrastructure?	MNO? Regulator?
Data Controller – Who decides on data usage for national statistics?	NSO?
Data Processor – Who develops methods and scripts/software?	MNO? NSO? Partner?
– Who executes scripts on the server?	MNO? NSO? Partner?

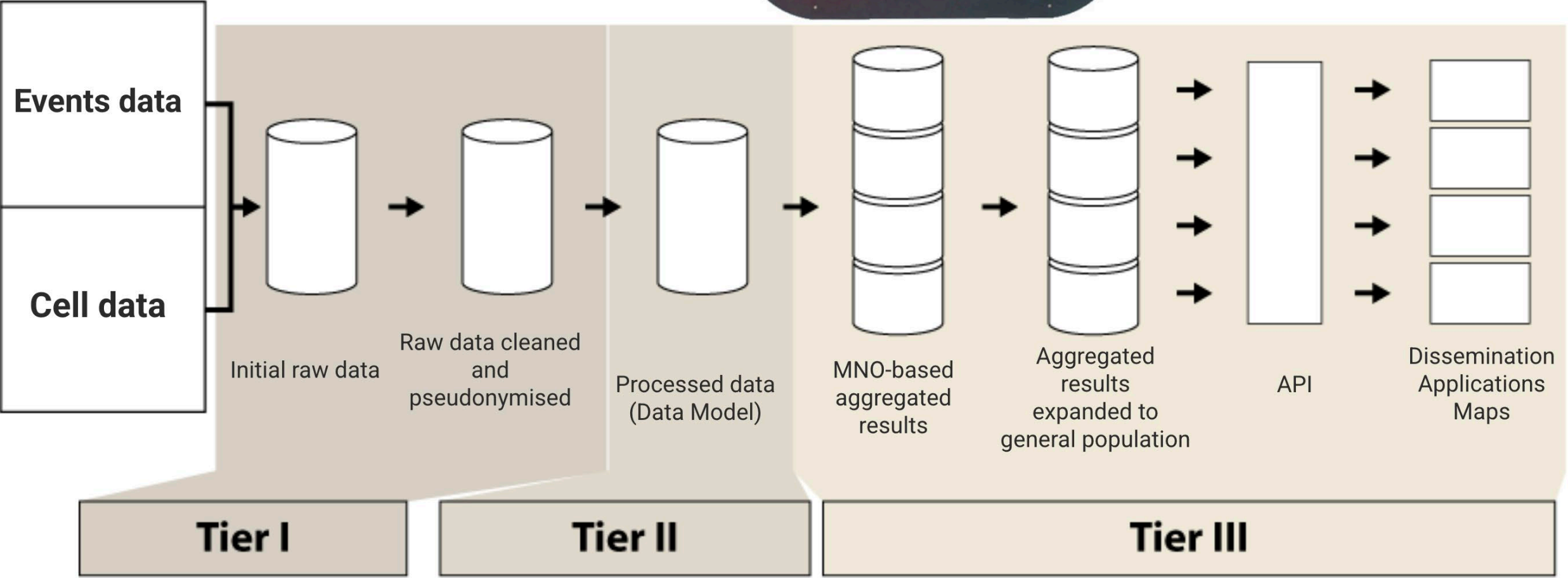
Decisions on data access modalities are determined by:

1.  Regulatory framework
2.  Stakeholders' preferences
3.  Stakeholders' capacity and mandate

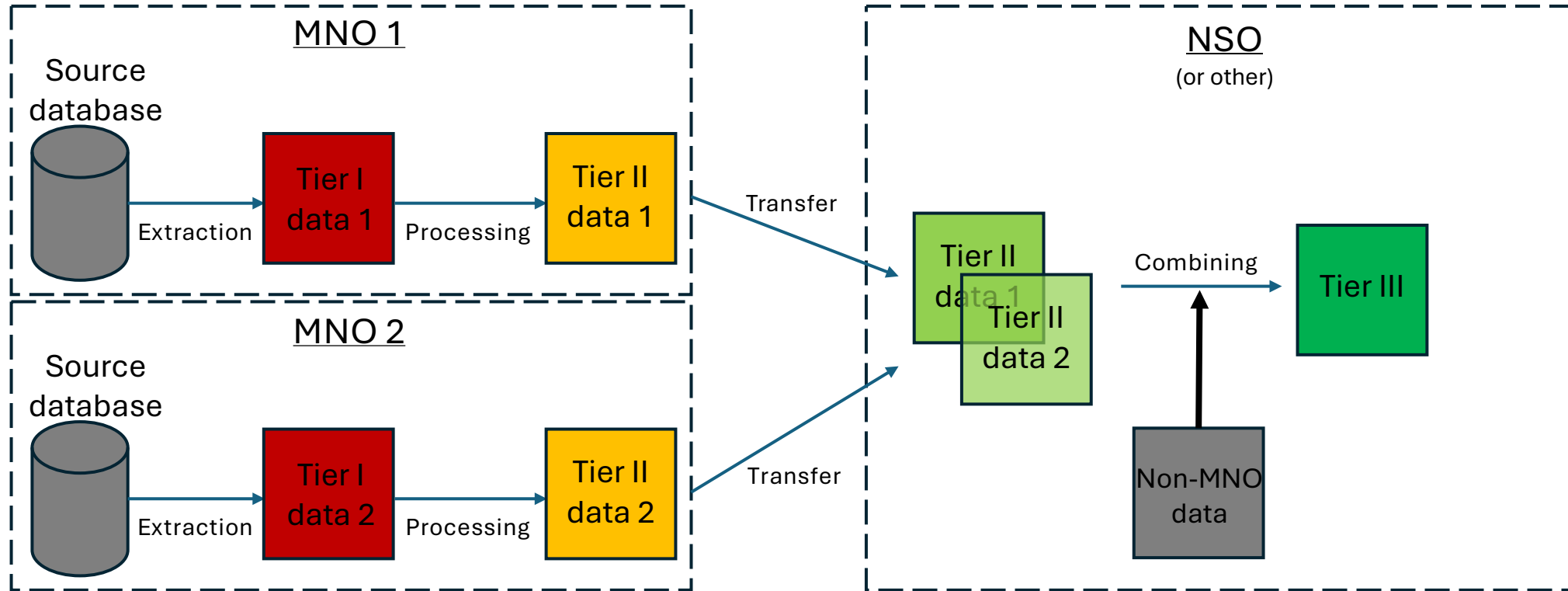
Overview of the main Data Access Models

Role	MNO-centered	Regulator-centered	Hybrid 1	Hybrid 2
Data owner	Always MNOs			
Data Custodian	MNO	Regulator	MNO	MNO
Data Controller	NSO	NSO	NSO	NSO
Data Processor (methods/scripts)	MNO	NSO/Partner	NSO/partner	NSO/partner
Data Processor (execution)	MNO	NSO/Partner	NSO/partner	MNO

3 Tiers for MPD Data

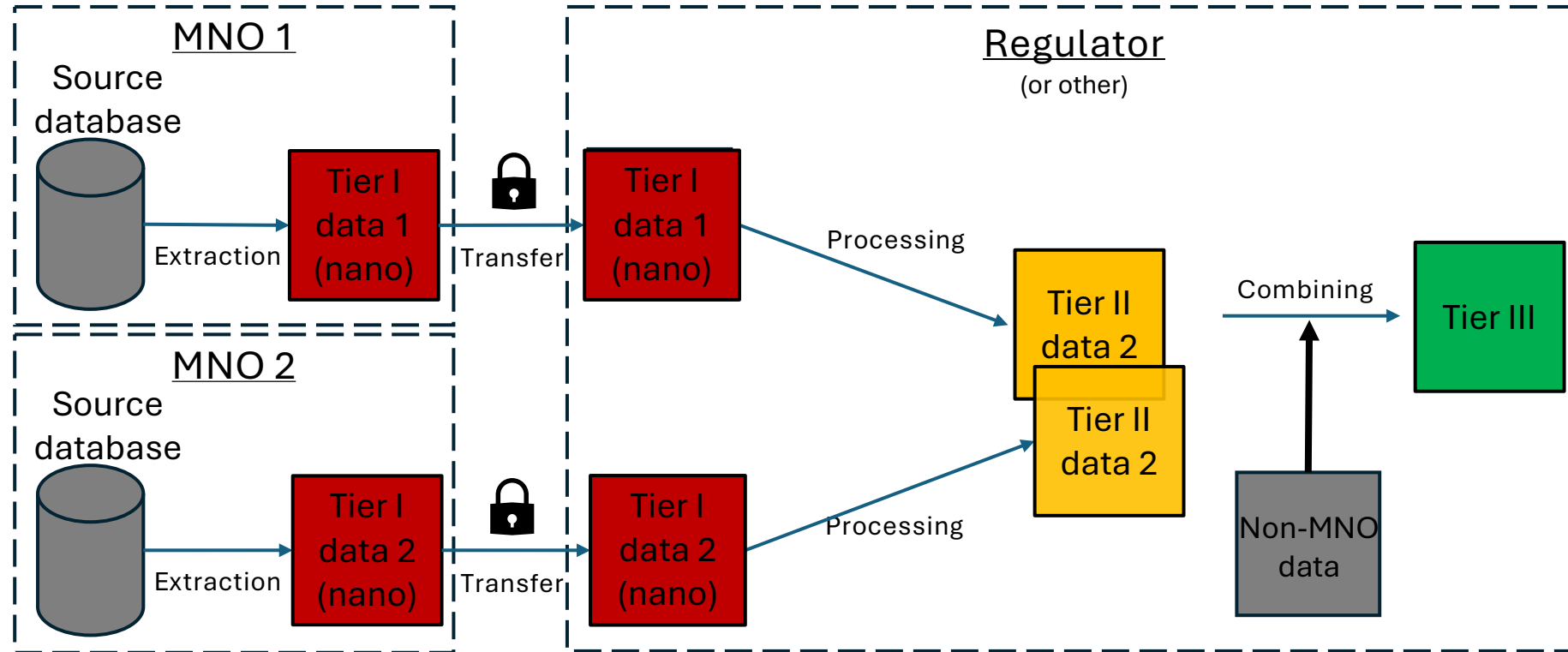


Data Pipeline Model A – “MNO-centered”



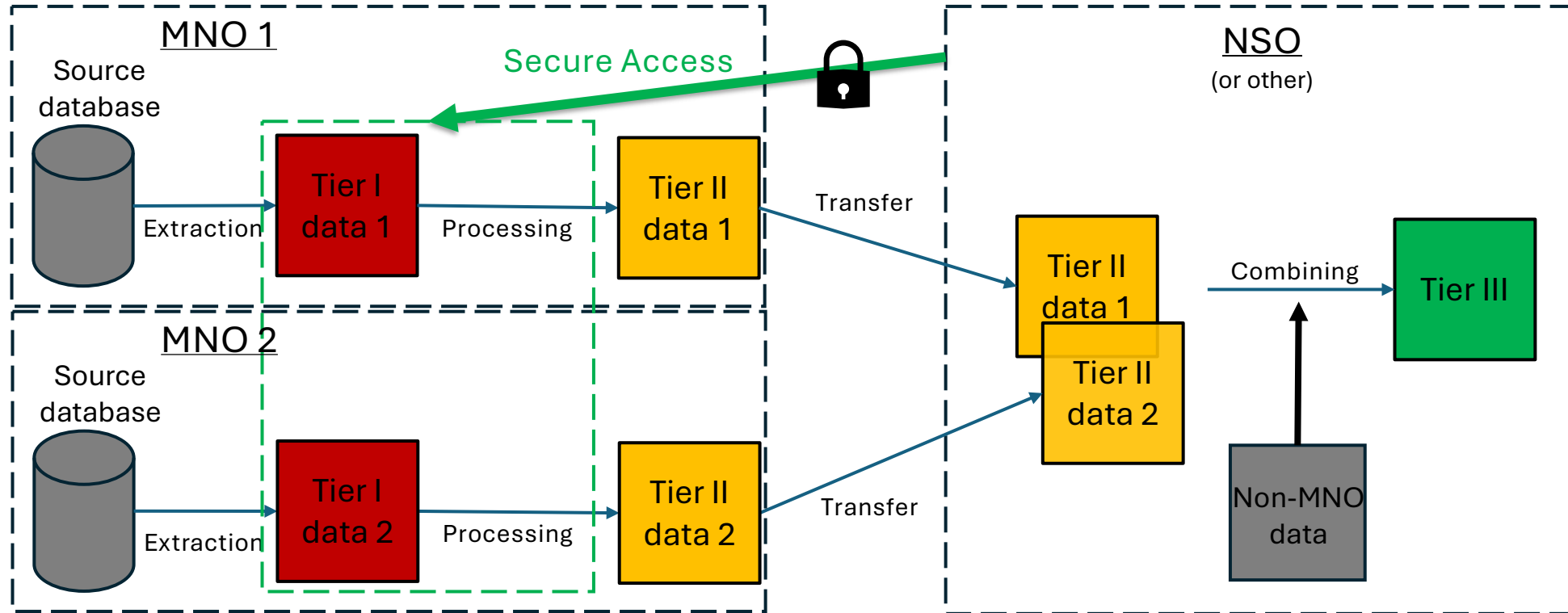
- Data stored on MNO infrastructure
- Tier I Raw data processed by MNO, i.e. no one but the MNO has direct access to raw CDR data.
- Tier II data shared with government entities.

Data Pipeline Model B – “*NSO/Regulator-centered*”



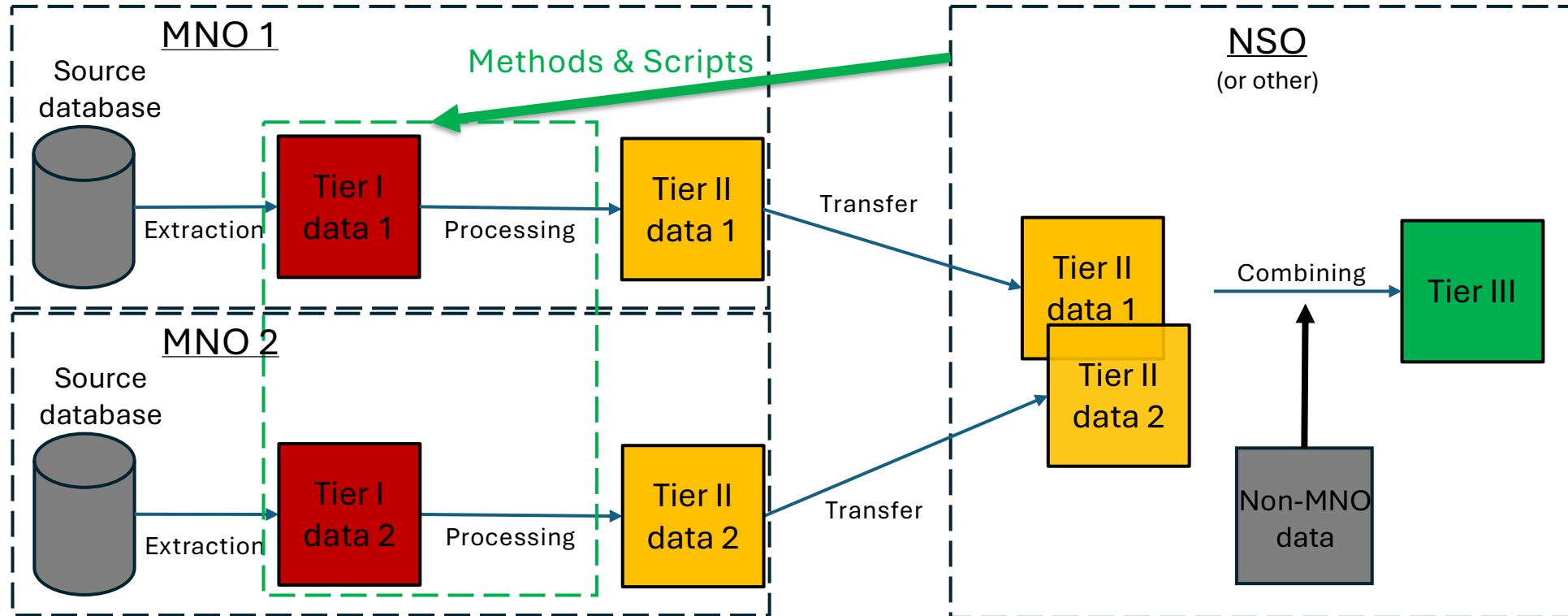
- **Legal mandate of ICT regulator to collect CDR data from MNOs**
- Data stored on regulator's infrastructure.
- Data processed on regulator's infrastructure by NSO, other gov. entity or technical partner.

Data Pipeline Model AB – Hybrid 1



- Tier I nano data remain on MNO's infrastructure.
- Data processed on MNO's infrastructure BUT government and technical partners can access the server for data processing (either on MNO's premises or via VPN).

Data Pipeline Model AB – Hybrid 2



- Tier I nano data remain on MNO's infrastructure.
- Data processed on MNO's infrastructure
- **BUT government and technical partners can provide scripts to be executed on the MNO's server.**
- Scripts are executed by the MNO | **no direct access to Tier I data** is provided.

From data access model to functioning data systems

- Data arrangements need to be **formally institutionalized** via dedicated legal tools:
 -  Clear privacy and security safeguards
 -  Data access conditions
 -  Clear rules and protocols for data access, processing, sharing.
- **Strong institutional foundations** ensure:
 -  Transparency and accountability, reinforcing trust
 -  Sustainability
 -  Efficiency
- Primary legal tool for institutionalizing data arrangements = **Memorandum of Understanding (MoU)**

Five key pillars in an MoU

1. Roles and Responsibilities

Who's responsible for what (data custodian, controller, processor...)?

2. Data Sharing Protocols

What data is shared and how?

3. Data Access Protocols

Who accesses the data and how?

4. Permitted Data Use and Dissemination

What indicators are permitted and how are results disseminated?

5. Data Safeguards

How are data security and privacy ensured throughout the data pipeline?

World Bank MoU templates as a useful resource

- MoU templates were developed and accommodate various data access models.
- Available in English and French.
- What MoU templates are useful:
 - ✓ As a well-structured starting point to support MoU negotiations
 - ✓ To support the design of required protocols
- Have already been used in many countries: The Gambia, Botswana...
- **MoU templates are not a one-size-fits-all resource: you should always adapt it to your specific country context.**



[GDF-MPD Knowledge Portal – download MoU templates](#)



GDF GLOBAL DATA FACILITY

Thank you!

Paul Blanchard (World Bank, DECD)

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Cathy Riley (Flowminder Foundation)

cathy.riley@flowminder.org



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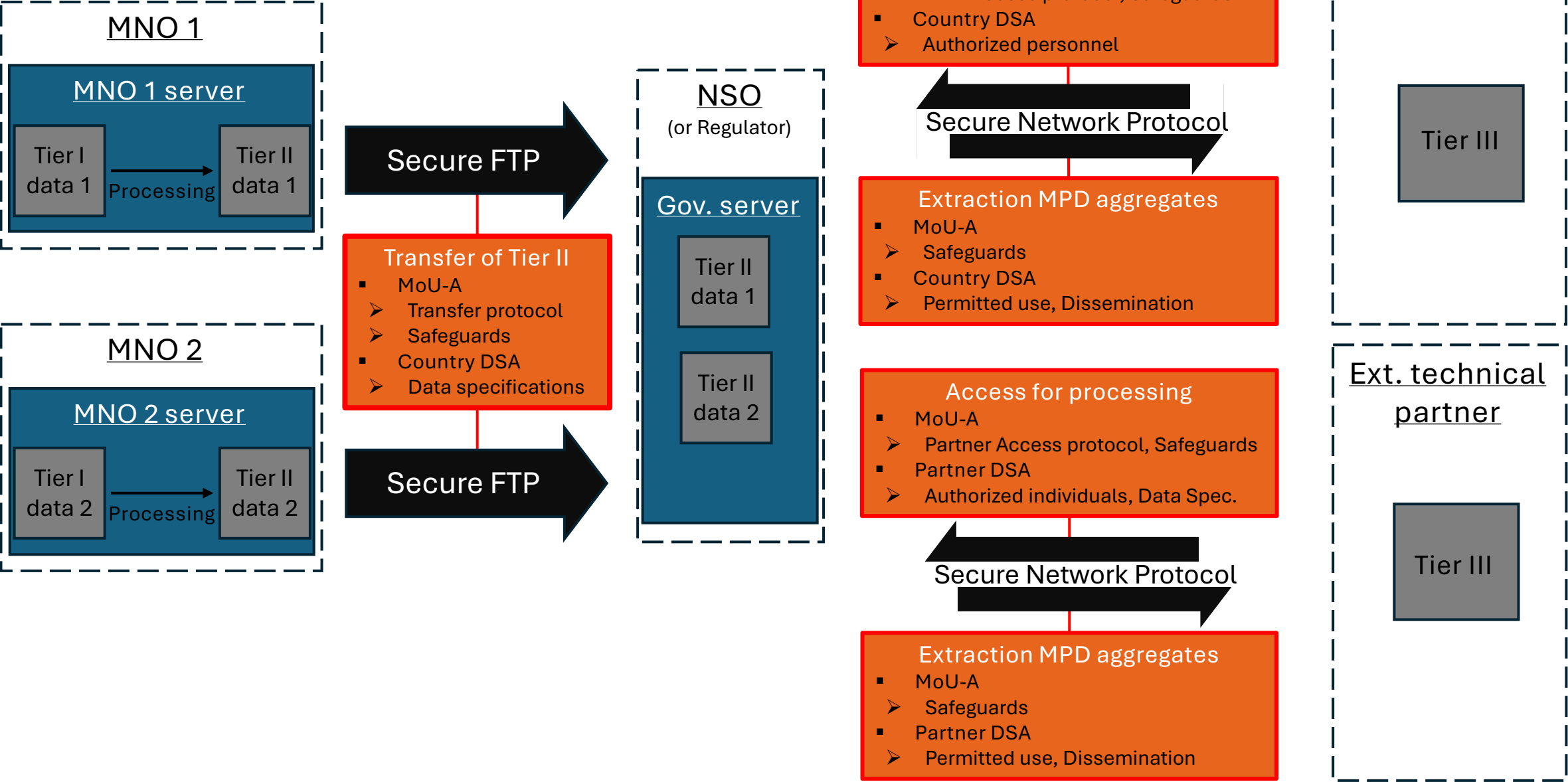
GOBIERNO
DE ESPAÑA

MINISTERIO
DE ECONOMÍA, COMERCIO
Y EMPRESA

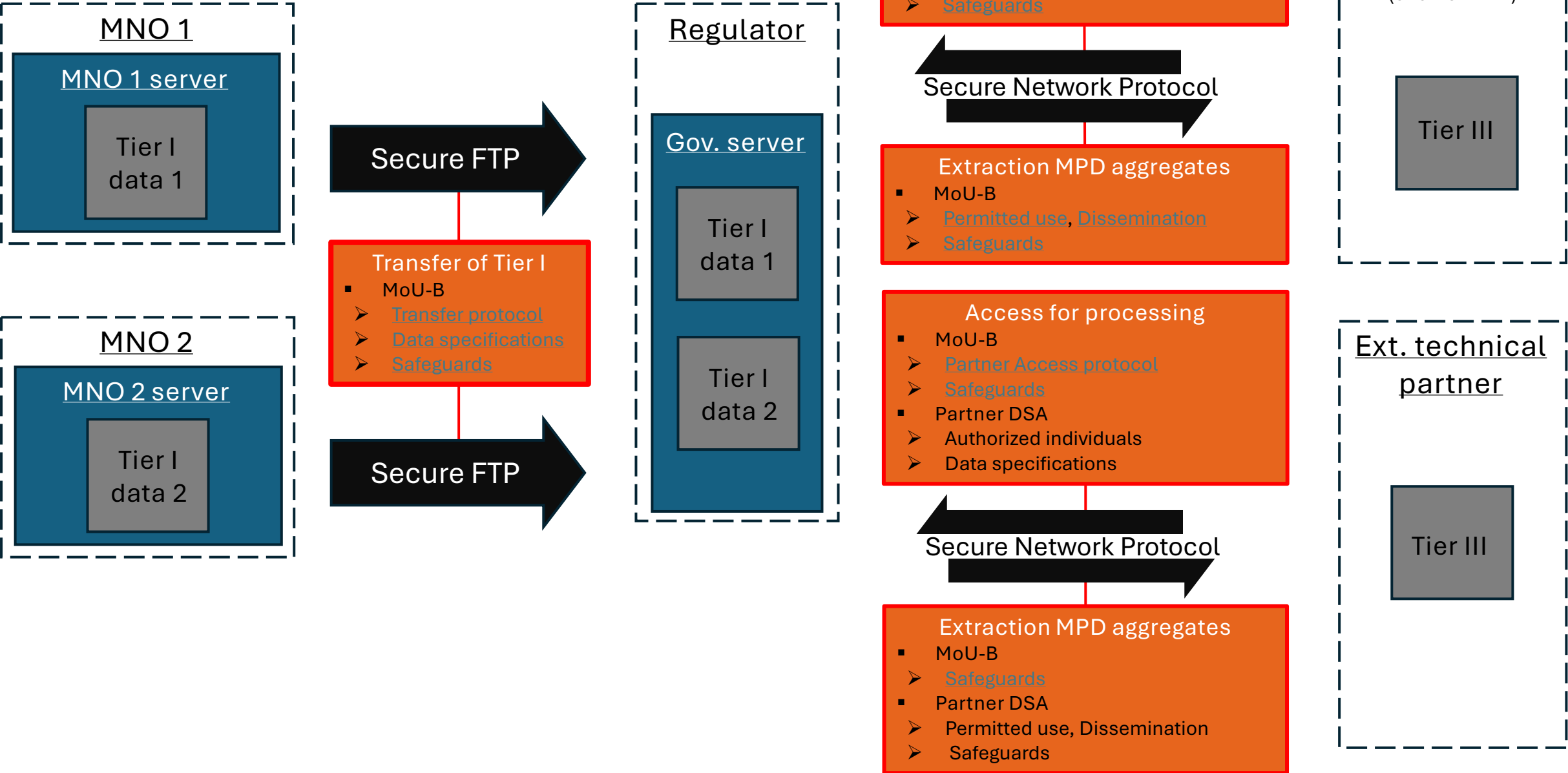


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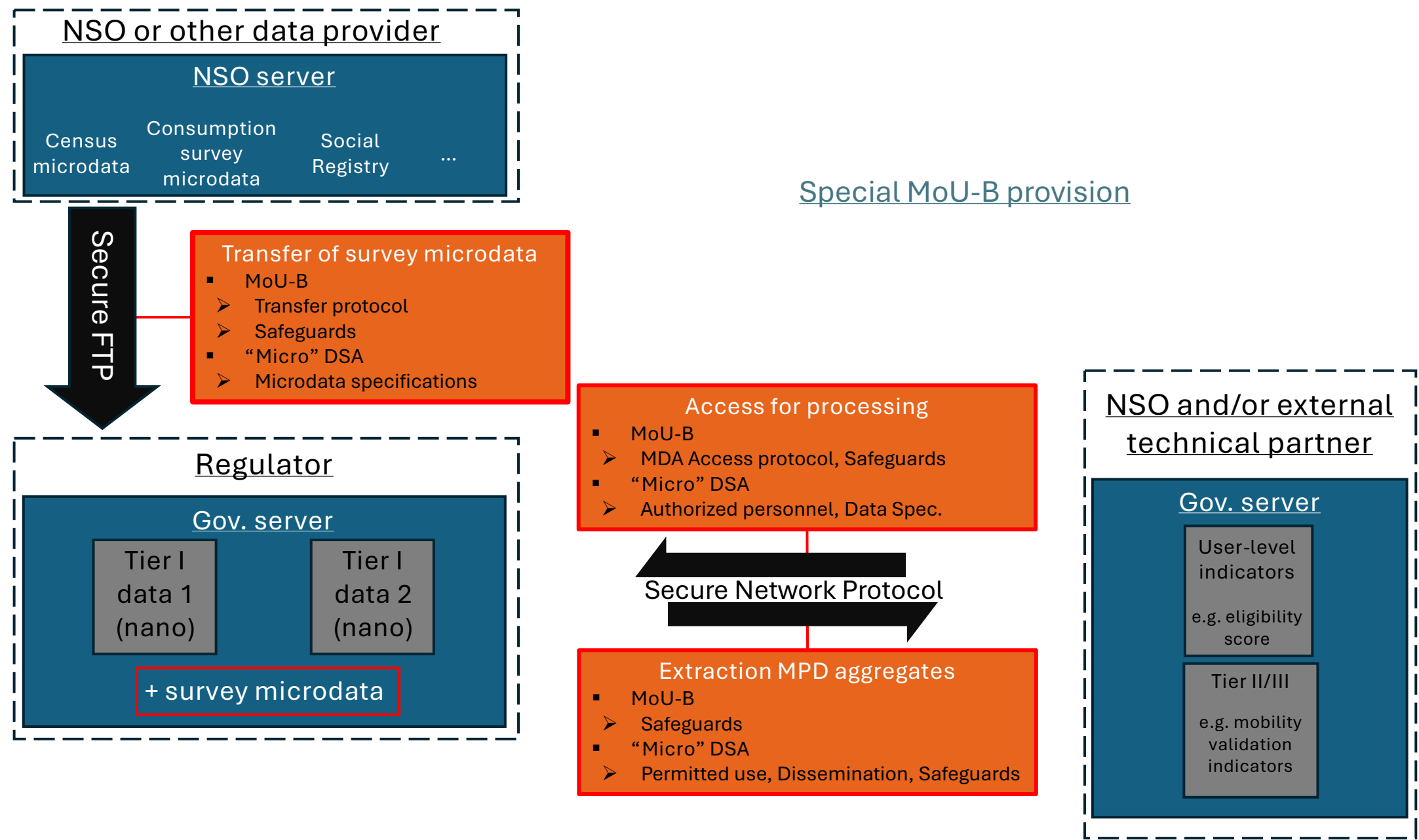
Institutionalization of Model A



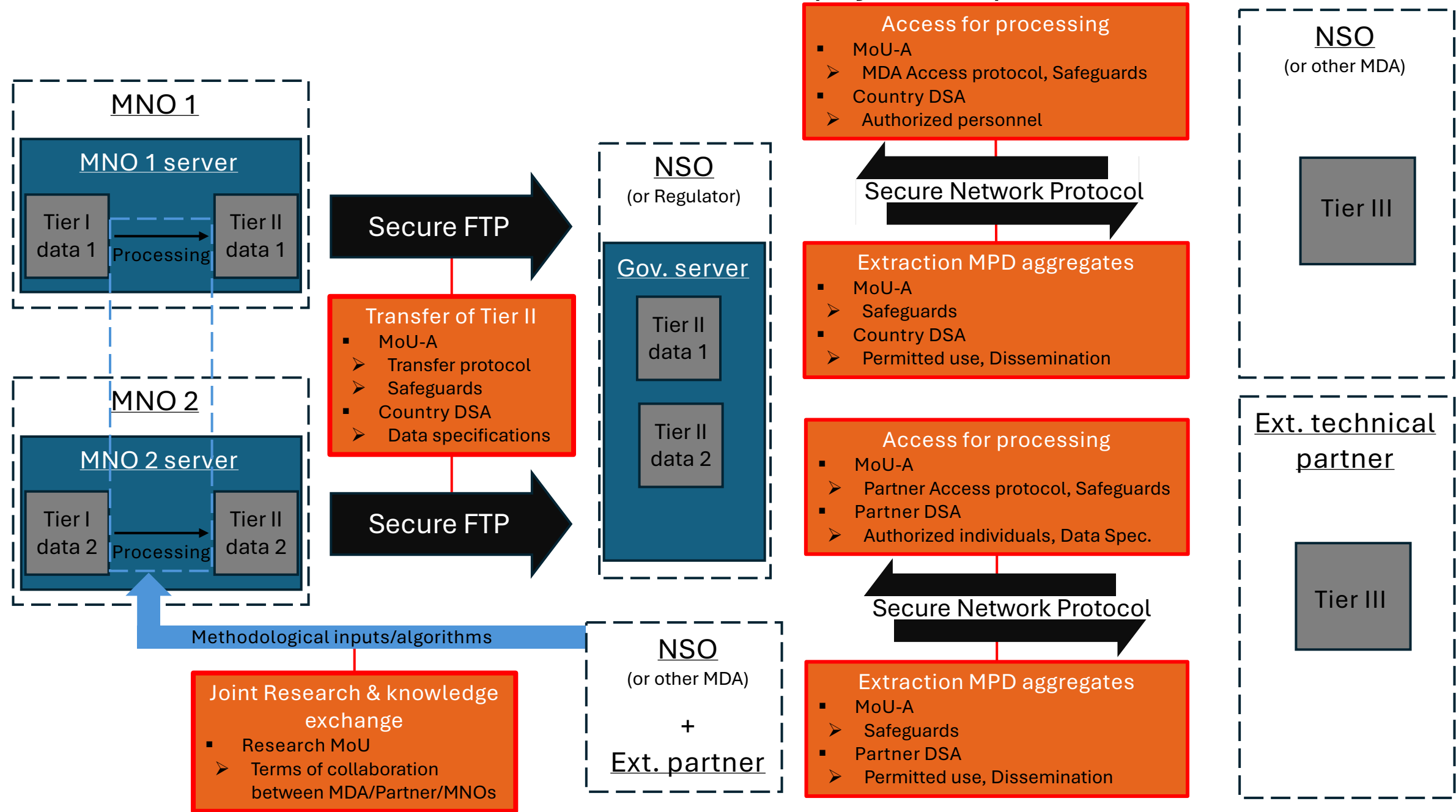
Institutionalization of Model B



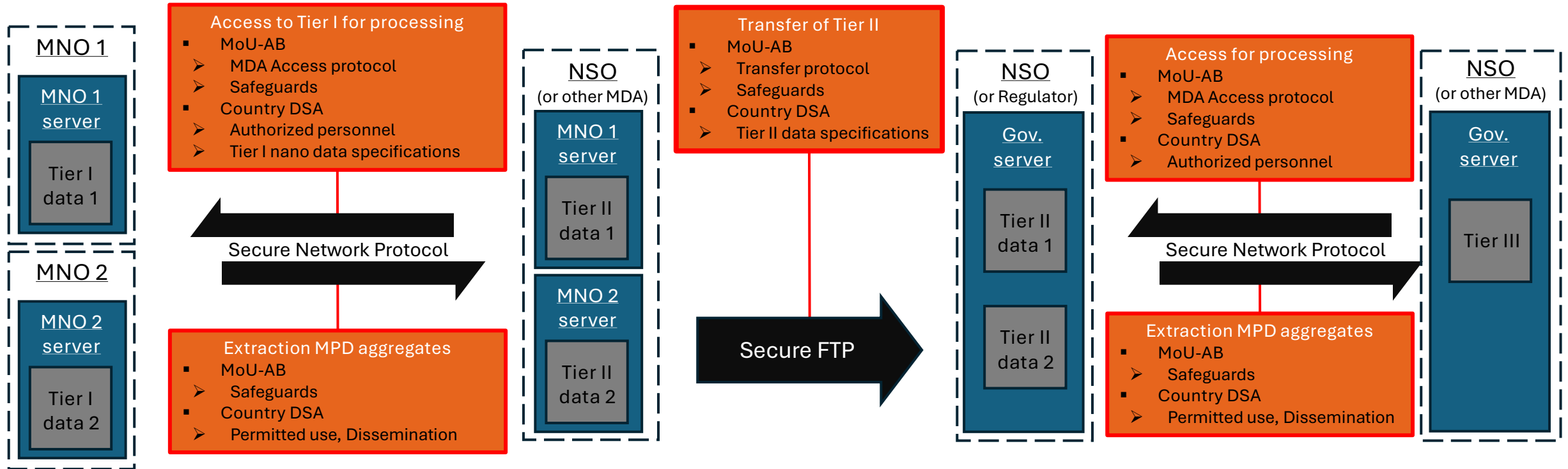
Institutionalization of Model B – combining phone records and survey microdata



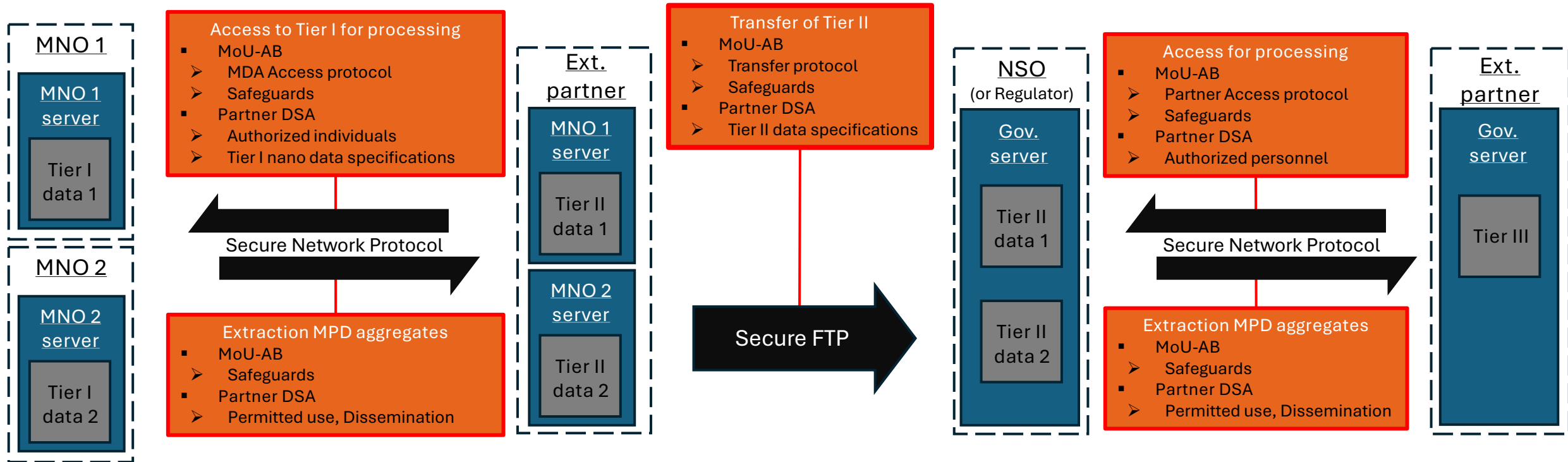
Institutionalization of Model A modified (Hybrid 2)



Institutionalization of Model AB – Government channel

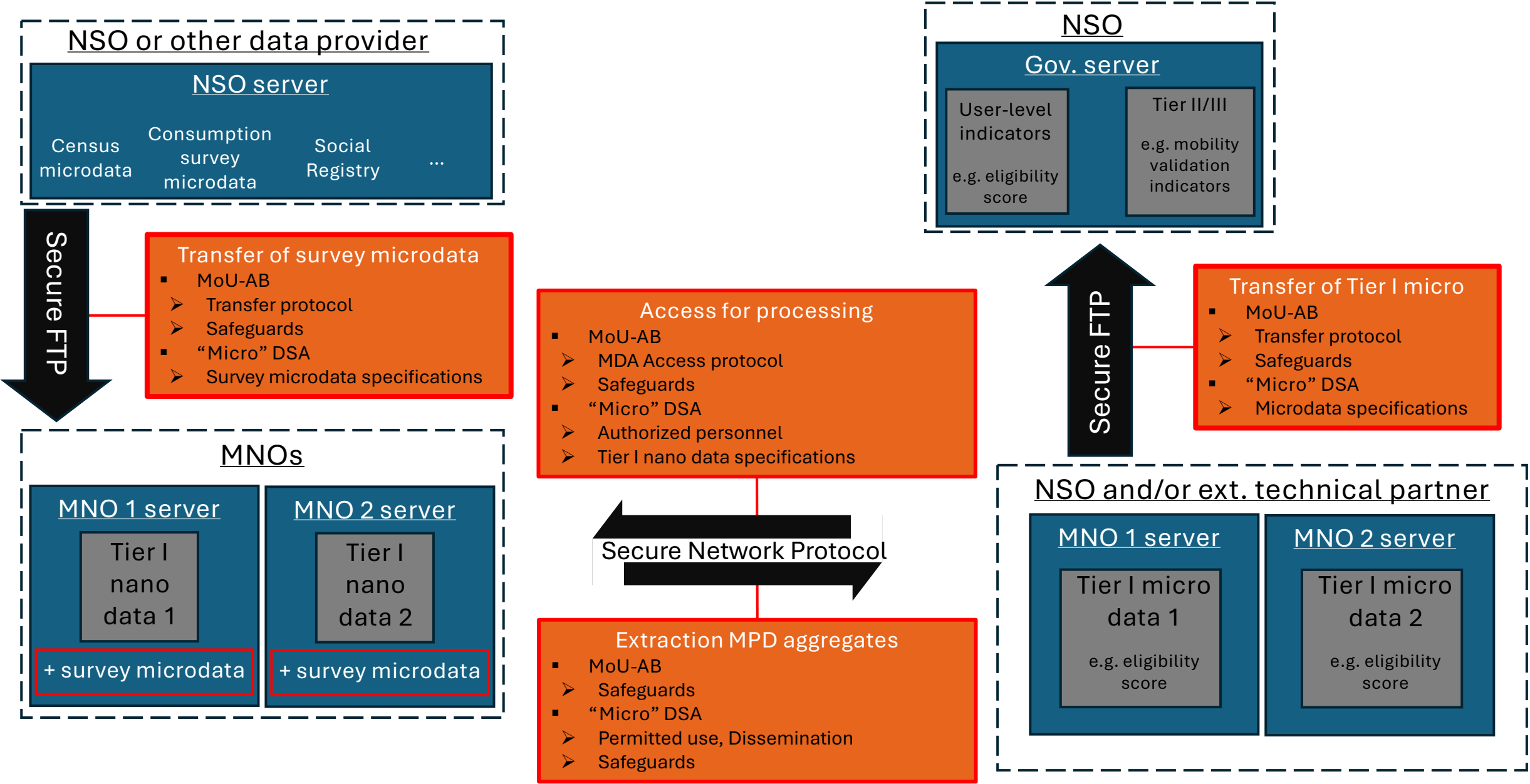


Institutionalization of Model AB – Technical.partner.channel



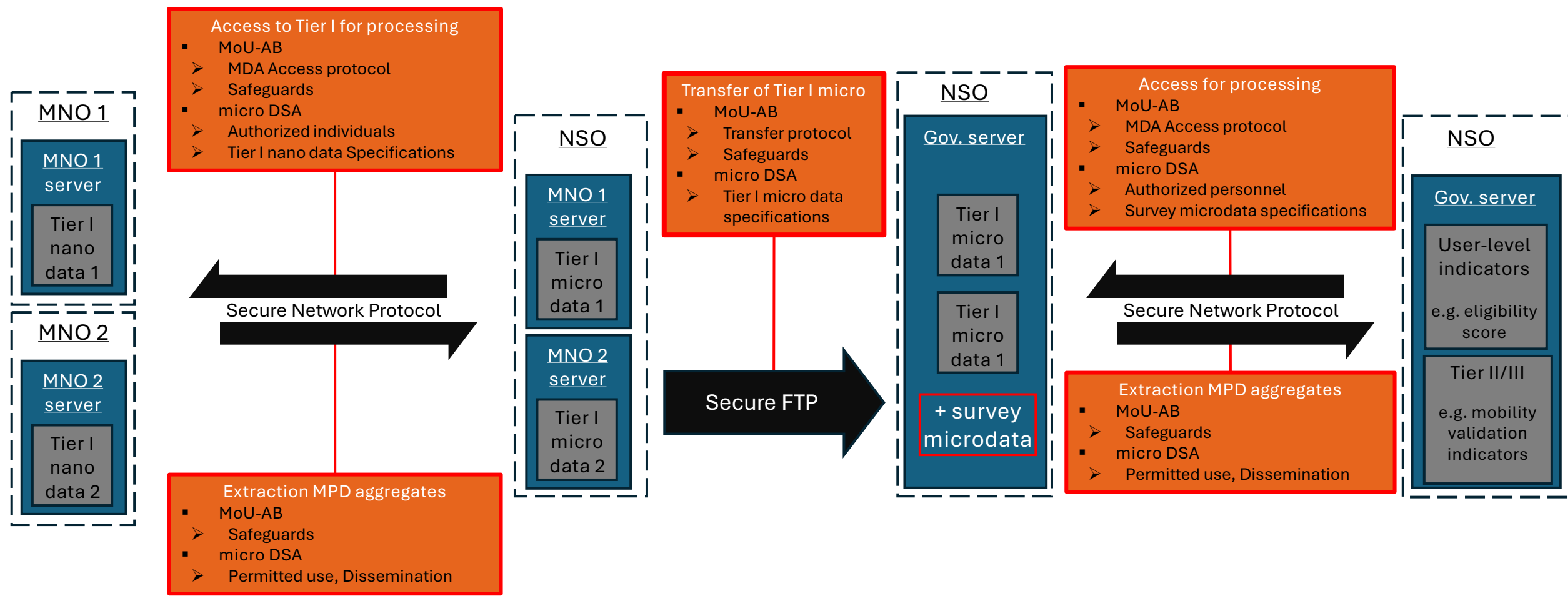
Institutionalization of Model AB – combining phone records and survey microdata

Option 1: on MNO server



Institutionalization of Model AB – combining.phone.records.and.survey.microdata

Option.8;on.NSO–government.server



MoU-B, clause 8 – Data Sharing Mechanism

8 Data Sharing Mechanism

- 8.1 Each participating MNO shall transfer Tier I nano data to [Regulator] in accordance with specifications provided in the Data Specifications Sheet, included in Appendix 1 of this Agreement. The data shall be transferred *[frequency of data transfer e.g. quarterly]*, specifically *[specify systematic transfer dates; for instance, if quarterly transfer: “on the first day of January, April, July, and October each year”]*. Each transfer shall cover the period from the day following the last day of the previous data transfer period up to the day preceding the current transfer date.
- 8.2 The Parties recognize that updates to the Data Specifications Sheet may be necessary in response to evolving operational needs. Any modifications to the Data Specifications Sheet shall be reviewed and approved through the appropriate governance channels, i.e. during a meeting of the *[Name of the coordination mechanism]*, ensuring compliance with applicable data protection and privacy regulations.
- 8.3 Each participating MNO shall transfer requested MNO data to [Regulator] via a secure file transfer protocol set up and maintained by [Regulator].
- 8.4 In addition to the regular data flow established by clause 7.1, [Regulator] may submit exceptional data requests to participating MNOs to address specific needs, such as supporting emergency response efforts. Such exceptional data transfers shall be governed by the submission of an Exceptional Data Request Form (EDRF). An EDRF template is provided in Appendix 2 of this Agreement. The EDRF shall provide technical specifications for the requested data, including data type, variables, and temporal coverage. MNOs shall commit to transfer requested data within *[X]* business days upon receipt of an EDRF.
- 8.5 Each data transfer shall be accompanied by an up-to-date file containing the unique identifier of every active antenna during the relevant period, along with its corresponding coordinates. Additionally, each participating MNO shall maintain a log file that is regularly updated to record all instances of antenna installations or removals on the mobile network infrastructure, along with the respective dates of these events.

What data? → Data Specifications Sheet

Appendix 1: Data Specifications Sheet governing regular MPD transfers¶

Type of data requested:¶

Example:¶

Call Detail Records (CDR) containing all Voice, SMS and data events¶

Variables:¶

Example:¶

- → International Mobile Equipment Identity (IMEI) of the caller (device identifier)¶
- → International Mobile Subscriber Identity (IMSI) of the caller (SIM card identifier)¶
- → Event start time¶
 - → Format: YYYY-MM-DD-HH24:mm:ss¶
- → Event duration¶
- → Antenna identifier for the caller¶
- → Normal Location Area Code (LAC)¶
- → Event type¶
 - → Voice, SMS, or Data¶
- → Technology¶
 - → 2G, 3G, 4G, LTE, 5G¶

When are data shared? → clause 8.1

Exceptional data requests → clause 8.4

MoU-B, clause 11 – Data Safeguards, transfer of Tier I

11 Data Safeguards

- 11.1 The Parties shall jointly define the Statistical Disclosure Control (SDC) methods to be applied (e.g. k-anonymity property) to ensure that any personally identifiable information is fully anonymized in final indicators intended for public disclosure, reducing the risk of disclosing information on statistical units and preventing any possibility of re-identification.
- 11.2 *[Regulator]* shall implement a pseudonymization procedure of the highest standard to ensure the privacy and security of Tier I data obtained from participating MNOs. This procedure shall comply with internationally recognized best practices and regulatory requirements for data protection. The *[Regulator]* shall also regularly review and update the pseudonymization process in response to technological advancements and evolving privacy risks to maintain the highest level of data protection.
- 11.3 The Parties shall adhere to the data minimization principle, ensuring that only data necessary and appropriate for the intended purpose is used, in alignment with the objectives set forth in this Agreement or as subsequently agreed upon by the Parties in the applicable data sharing agreement.
- 11.4 The Parties shall strictly adhere to data sharing, access, processing and disclosure protocols established under clauses 8, 9, and 10, ensuring transparency and accountability on the storage location of MNO data throughout all stages of the processing pipeline, the individuals granted access and the intended purpose of data use, altogether minimizing the risk of data breaches and misuse.
- 11.5 *[Regulator]* and *[MNO 1 or MNOs]* shall jointly implement appropriate encryption measures to enhance data security during the transfer of Tier 1 nano data from *[MNO 1 or MNOs]* to *[Regulator]*' server.

MoU-B, clause 9.1 – Data Safeguards, data access

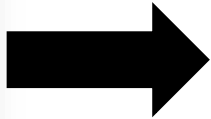
9 Data Access

- 9.1 Tier I data shared by MNOs and stored on *[Regulator]*'s server will be accessible and processed via a secure network protocol such as a Secure Shell connection (SSH), which will be set up and maintained by *[Regulator]*.
- 9.2 A list of authorized personnel, employed by one of the Parties, shall be granted access to *[Regulator]*'s server to access and process Tier I data using the designated secure network protocol. Each authorized individual will be assigned unique credentials. The list of individuals approved for data access is provided in Appendix 3 of this Agreement.
- 9.3 The list of individuals pre-authorized for data access may be modified, subject to mutual agreement by all Parties.

MoU-B, clause 9.2 – Data Access for NSO

9 Data Access

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Who may access Tier I data?
→ List of authorized personnel

Appendix 3: list of pre-authorized individuals for access to *[Regulator]*'s server

First name	Last name	Institution	Job title	Access restriction
...	...	...	...	...

[Back](#)

MoU-B, clause 10 – Dissemination of results

- 10.4 The Parties agree that any research results or statistical findings produced under this MoU can be published, provided that all Parties have been properly credited, and any proprietary or sensitive information has been appropriately anonymized.
- 10.5 Before any publication is released to the public, all parties will be notified in writing and granted **[X]** days to review the content. The absence of a response after this period shall be deemed as non-objection and the publication shall be considered approved by all Parties. Any disputes concerning publication will be resolved by mutual agreement or arbitration.
- 10.6 Indicators and results derived from MNO data that The Parties deem too sensitive for public disclosure but are intended for sharing only with a specific Government of **[Country Name]** Ministry, Department or Agency (MDA) may be approved for restricted disclosure. These results may be shared exclusively with the designated MDA, provided that the MDA signs a mutual Non-Disclosure Agreement with The Parties, committing to maintain confidentiality and refrain from further dissemination.

MoU-B, clause 11 – Data Safeguards, extraction

11 Data Safeguards

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- 11.3 The Parties shall adhere to the data minimization principle, ensuring that only data necessary and appropriate for the intended purpose is used, in alignment with the objectives set forth in this Agreement or as subsequently agreed upon by the Parties in the applicable data sharing agreement.
- 11.4 The Parties shall strictly adhere to data sharing, access, processing and disclosure protocols established under clauses 8, 9, and 10, ensuring transparency and accountability on the storage location of MNO data throughout all stages of the processing pipeline, the individuals granted access and the intended purpose of data use, altogether minimizing the risk of data breaches and misuse.
- 11.5 *[Regulator]* and *[MNO 1 or MNOs]* shall jointly implement appropriate encryption measures to enhance data security during the transfer of Tier 1 nano data from *[MNO 1 or MNOs]* to *[Regulator]*'server.



9.4 *[This clause governs access to Tier I/II data stored on the Regulator's server for external partners. The template proposes two implementation options.]*

Given the potential need for technical expertise, *[Regulator]* and *[NSO]* may establish future commitments or seek external collaborations with relevant institutions. Such arrangements shall be formalized through separate agreements, subject to applicable data governance and confidentiality provisions. Consequently, external individuals, including researchers and technical partners, may be granted access to *[Regulator]*'s server for the purpose of accessing and processing Tier I or Tier II MNO data under the following requirements:

9.4.1 *[Option 1: access to external partners requires approval by all Parties of the MoU]* One Party shall submit a draft "Partner" Data Sharing Agreement (PDSA) to the other Parties, who must respond within *[X]* days of receipt with either a signed agreement, proposed amendments for approval, or a rejection. The PDSA must be approved and signed by all Parties to authorize data access and processing.

[Option 2: Regulator granted full mandate to manage and regulate access for external partners to MNO data stored on their server.] *[Regulator]* shall have the full authority to evaluate, approve, and manage access for external partners to MNO data stored on *[Regulator]*'s server, including researchers and technical collaborators, in alignment with the objectives of this MoU. Such access shall be granted only upon the signing of a dedicated "Partner" Data Sharing Agreement (PDSA) between *[NSO]*, *[Regulator]* and the external partner.

9.4.2 The PDSA must specify the terms of data access, including any data access restrictions, the duration of access, and the intended purpose of data use.

9.4.3 Upon signing a PDSA, *[Regulator]* shall provide the requesting entity with the necessary credentials to perform their duties. If a specific access duration is defined, *[Regulator]* shall ensure that the credentials issued under the PDSA are revoked upon its expiration.

9.4.4 The intended data use must not involve pairing Tier I MNO data with individual- or household-level survey information. Data access for such applications shall be regulated by separate "micro" Data Sharing Agreement under the terms and conditions outlined in clause 10.2.



How external partners may access Tier I data?

→ Partner Data Sharing Agreement (PDSA)

- PDSA = channel via which external partners may access individual-level phone records.

[Back](#)

MoU-B, clause 10.7 – combining survey microdata with phone records

- 10.7 Applications involving the pairing of Tier I MNO data with individual- or household-level survey information – e.g. for the application of bias correction methods – shall require a “micro” Data Sharing Agreement (micro DSA) that meets conditions outlined in clause 9.4. A micro DSA shall explicitly state the pairing of MNO data with survey microdata as part of the intended data use and shall specify the survey microdata to be used for the intended application.
- 10.7.1 In the framework of a micro DSA, *[NSO]* shall facilitate the secure transfer of the required survey microdata to *[Regulator]*’s server via the designated secure file transfer protocol.
- 10.7.2 *[NSO]* shall apply an appropriate pseudonymization procedure to ensure both user anonymity and alignment with the pseudonymization of Tier I MNO data, thereby enabling effective pairing of both data sources.
- 10.7.3 *[Regulator]* shall ensure the protection of survey microdata stored on its server by enforcing restrictions on access, transfer, download and modification. Specifically, access to designated survey microdata on *[Regulator]*’s server shall be strictly limited to individuals listed in a signed micro DSA that explicitly states the use of such survey microdata in its associated intended data use.
- 10.7.4 In the framework of applications involving the pairing of individual phone records with individual- or household-level survey information, any resulting indicators intended for publication shall comply with privacy regulations and adopt appropriate measures to maintain user anonymity.



Mobile Phone Data for Policy Program



gdfmpd@worldbank.org



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+

Top Statical Applications of Mobile Phone Data

<https://unstats.un.org/bigdata/task-teams/mobile-phone/index.cshtml>



UN MPD Task Team



Tourism
statistics

1



Social
Assitance

2



Dynamic
population

3



Displacement,
Migration,
Crisis
Response

4



Information
society
indicators

5



Transport
and commuting
statistics

6



+

MPD Statistics: Demonstrated Value

Estonia

4x faster processing

2.5x more cost-efficient

200x larger sample size than traditional surveys

12x more geographic breakdown (country-level)



Indonesia

Cost savings: From **\$2.5M** (traditional) to **\$950K** (MPD-based)

Savings of **~\$1.55M USD** per round

Higher spatial & temporal resolution (monthly, quarterly, yearly)

=



Spain

Improved timeliness: Tourism results published within 1 month after reference month

Larger sample sizes, better data quality (reduced recall bias and data entry errors)

Reduced burden on respondents

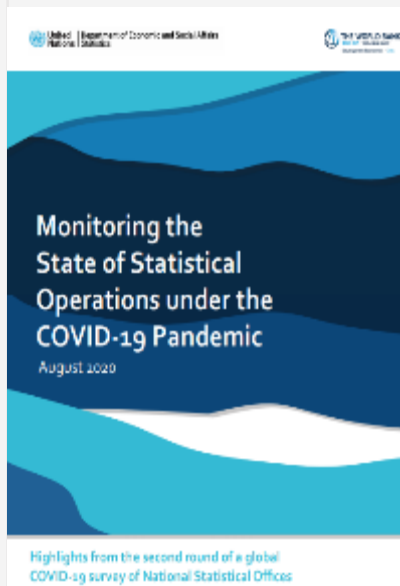


MPD Capacity is a Priority for NSOs

National Statistical Office Priorities for Capacity Development

Question

In the next three years, does your NSO want to expand capacity in the following areas to respond to the new data demands of the 2030 Agenda for sustainable development/SDGs and more recently the COVID-19 pandemic?

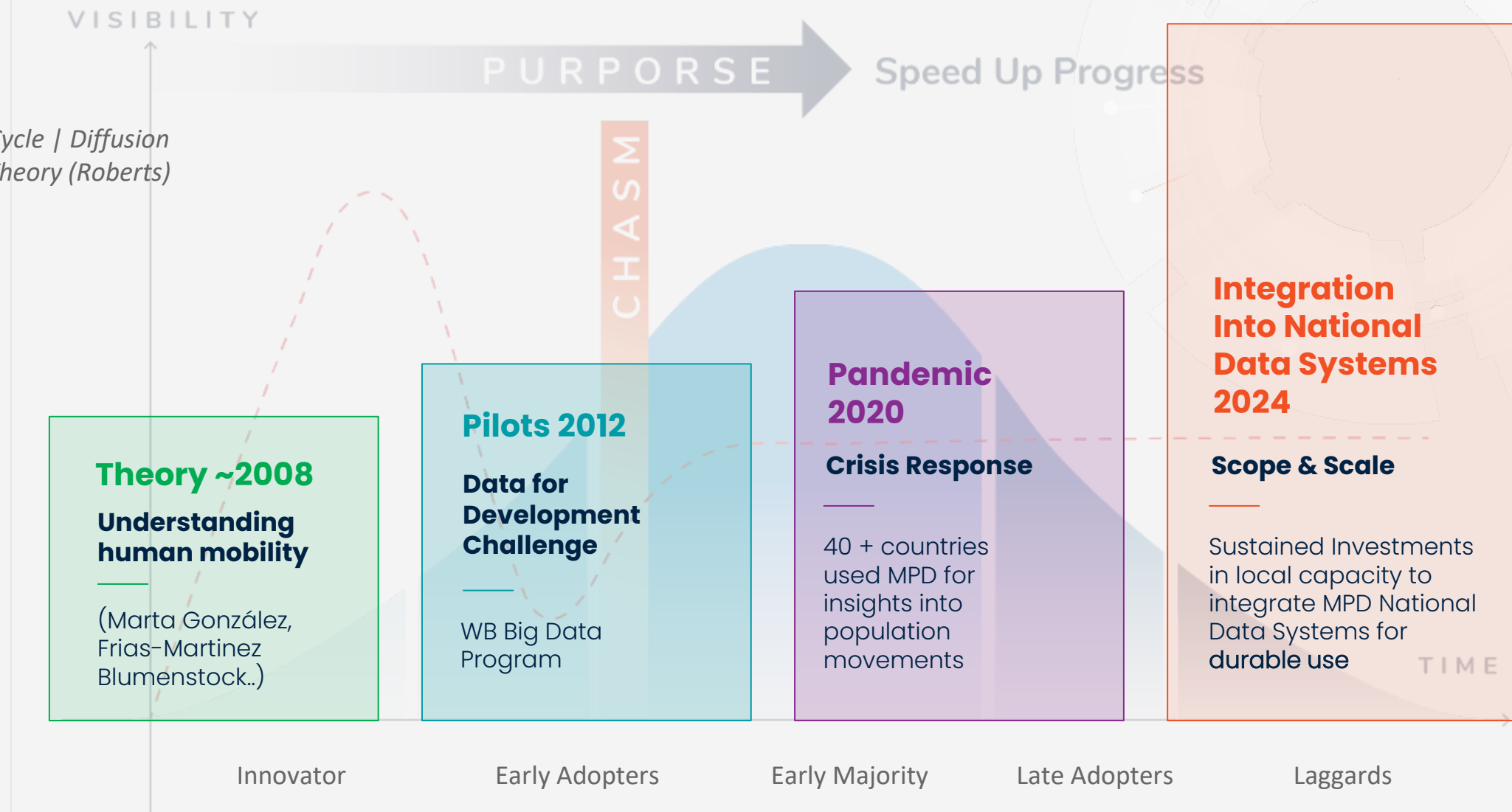


78% of National Statistical Offices

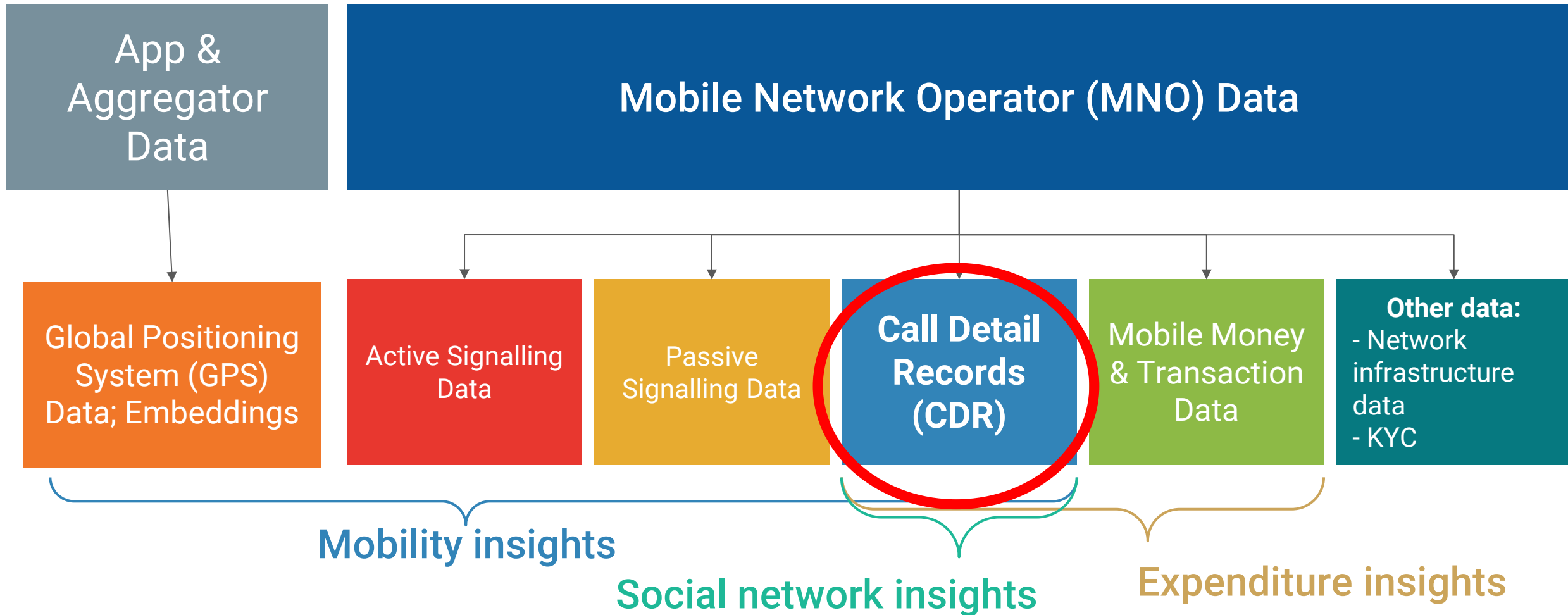
consider Mobile Big Data a priority over the next three years (CTAP Round 1)

Less than 20% of NSOs have capacity to use sophisticated data for Policy (World Development Report)

Evolution of **MPD for Policy**



Mobile Phone Data (MPD)





GDF-MPD Window and Cohort – Based Support Program

Global Data Facility– Mobile Phone Data Program

Goal

Deliver programmatic support to integrate **Mobile Phone Data** into the national data systems of **30 countries by 2030**

Operations Pillar (DEC, Pov, Digital Transformation, ITU)

- **Supports Cohorts of Country Teams** to implement country activities to put MPD to work for Policy Applications and Statistics
- **MPD Foundations Training Program:** practical and technical workshops and online training programs on the Foundations of MPD for Policy
- **MPD Essential Resources:** Repository of public goods, including frameworks, tools, use cases, guidelines, content, etc

Research Pillar (DIME, External Board)

- Innovation in methods and standards at scale
- World class Research Board (Blumenstock, Frias-Martinez, Gonzalez)

Impact

- Supporting Largest Country Cohort to Operationalize MPD for Policy– 18+ Countries
- GDF-MPD grants used to amplify IDA/IBRD operations toward **Integrated National Data Systems**



Key stakeholders: Mobile Phone Data Country Ecosystem

Data Owners

- Mobile Network Operator(s)

Data Users

- National Statistical Offices
- Other Government Departments
- International Organisations
- Non-Governmental Organisations
- Private Sector

Regulators (Data Controllers)

- Data Protection Authority
- Telecommunications Regulator
- National Security Agencies

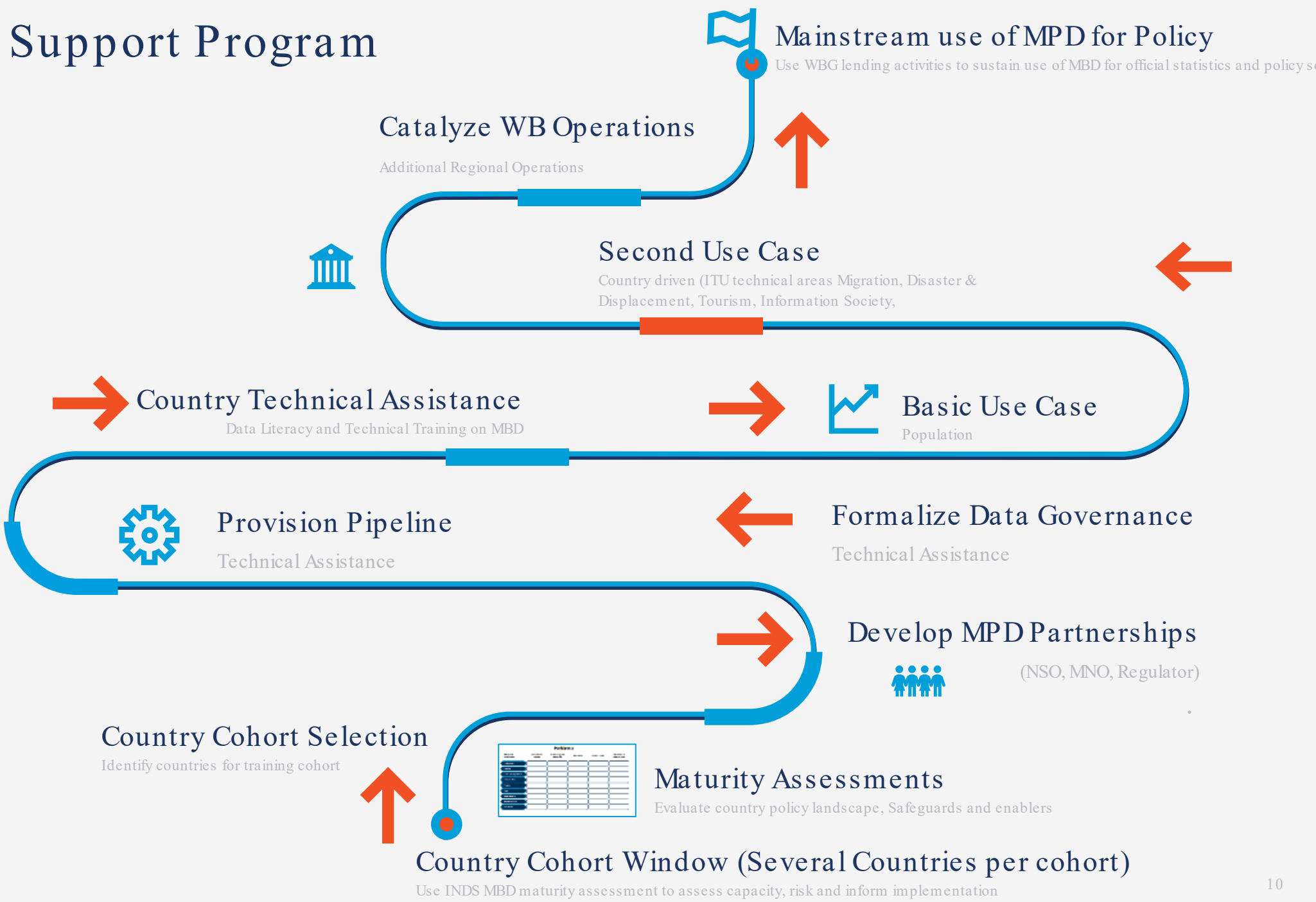
Technical Service Providers

Cohort-Based Support Program

Practitioner | Expert
Enablers, Standards and Safeguards

Foundation
Enablers, Standards and Safeguards

Pre-Foundation
Awareness Raising



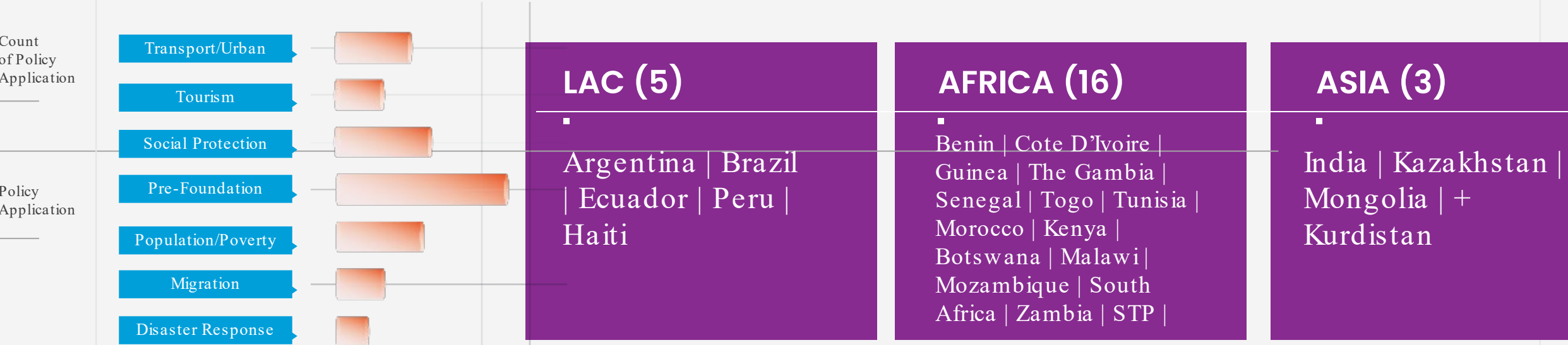
Cohort 1: 18 Country Teams



Cohort 2: 6 Country Teams (SADC)



At a Glance – Policy Areas



Active and Planned Country Cohorts

Cohort 1 (18 Country Teams)

Implementation Period

MPD Foundations
Training

Roll-over unused top-up funds

Cohort 2 (Subset of 16 Country Teams – Foundation Grants)

Implementation Period

MPD Foundations
Training

Cohort 3 (Planned)

Implementation Period

MPD Foundations Training

Cohort 4 (Planned)

Implementation Period

MPD Foundations
Training

2024

2025

2026

2027

2028

2029

Tools and Resources

Theory of Change as a planning tool

Putting Mobile Phone Data to work for Policy: Theory of Change Narrative

A summary narrative document to support readers of the Theory of Change for the Putting Mobile Phone Data to work for Policy programme hosted by the World Bank

Introduction

This document sets out in more detail the Theory of Change described [here](#) for the World Bank's programme 'Putting Mobile Phone Data to work for Policy' which is housed within the Global Data Facility (GDF), under a Mobile Phone Data (MPD) window. This Theory of Change is a work in progress and currently being consulted on (June 2024).

Background to the programme

Launched in 2023, the programme aims to 'mobilise investments to accelerate local capacity to integrate the responsible use of anonymized, aggregated MPD into National Data systems for real-time statistics and responsive policy solutions'. To read an explanatory powerpoint presentation about the programme, please see [here](#).

Why mobile phone data?

In LMICs with constraints to generating reliable data through traditional methods, and scarcity of data sources, non-traditional data and Big Data such as mobile phone metadata, can be applied and combined with other data, using new methods, to rapidly generate current information. Its strengths include that it has large geographic scale and penetration and high granularity, and that it is automatically generated in near-real time.

What are the barriers to use of mobile phone data?

The programme is being designed, in part, to tackle some of the barriers to adoption of mobile phone data being used as one of the tools that can inform data-driven decision making.

Some of these barriers to use include:

- low awareness and understanding of the potential, and limitations, of the data
- concerns about data protection for privately-held data
- limited experience of establishing public-private data partnerships for accessing privately-held data, and legal and data protection agreements
- cost of establishing data pipelines
- need for engineering expertise to establish reliable privacy protecting data pipelines and software
- scarcity of analytical skills to develop robust methods for different applications
- capacity gaps in LMICs, from technical capabilities to available human resources
- few fully realised and evidenced, replicable data use cases, and limited range of applications to date



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1 | 8

The Theory of Change

Background

Purpose

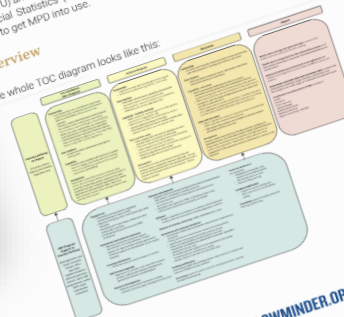
- Describe the shared vision of the programme
- Identify what needs to happen at country level, identifying key assumptions, and what the GDF-MPD programme needs to do, to support countries
- Provide a communication tool for sharing these ideas with relevant stakeholders for refinement and validation
- To guide monitoring and evaluation (the Theory of Change will be updated ad hoc, based on ongoing learning)

Development process

This Theory of Change was developed with the World Bank project team working on the Global Data Facility Mobile Phone Data window, together with staff from the International Telecommunications Union (ITU) and in consultation with members of the UN Committee of Experts on Big Data and Data for Official Statistics (UN-CEOD), drawing on collective experience in the sector regarding 'what works' to get MPD into use.

Overview

The whole TOC diagram looks like this:



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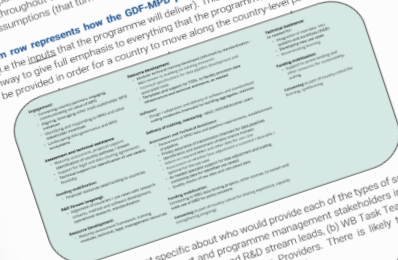
2 | 8

The **top row** of this diagram represents a **country-level pathway to a set of desired outcomes** including:

- A public-private **partnership** between country stakeholders
- Sustained, privacy protecting **provision of mobile phone data (MPD)**
- Stakeholder **capability to process and analyse** the data
- The **data being applied** to a national policy/programme related use case

This is intended to be a generic pathway that would be relevant to any country included in a WB-GDF programme cohort, and it does not presuppose a particular use case. Countries may be starting from different points in the pathway (maturity), stakeholders will vary; the partnership arrangements are likely to be different, and aspects such as how data is accessed, what types of data, who analyses and who uses it will vary across countries. But there are general similarities and some standard components that are required. Part of the purpose of developing and using the TOC as a learning tool throughout the programme is to identify the range of variation and any unanticipated actions, or assumptions (that turn out to be necessary conditions) for outcomes to be achieved.

The **bottom row** represents how the GDF-MPD program would support the country along the pathway (i.e. the **enablers** that the programme will deliver). This row is separated out from the country-level pathway to give full emphasis to everything that the programme team understands must be in place or be provided in order for a country to move along the country-level pathway.



This 'support pathway' is not specific about who would provide each of the types of support needed. For this programme, the key support and programme management stakeholders include: (a) GDF-MPD programme management core team and R&D stream leads, (b) WB Task Team Leads (TTLs), and (c) other suppliers, such as Technical Service Providers. There is likely to be variation in particular in how the specialist technical support is provided.

The support needs will probably also vary by country according to, for example, the stage of maturity (where they are starting from with partnerships and capabilities), and what kind and complexity of data analysis and use they want to deliver.

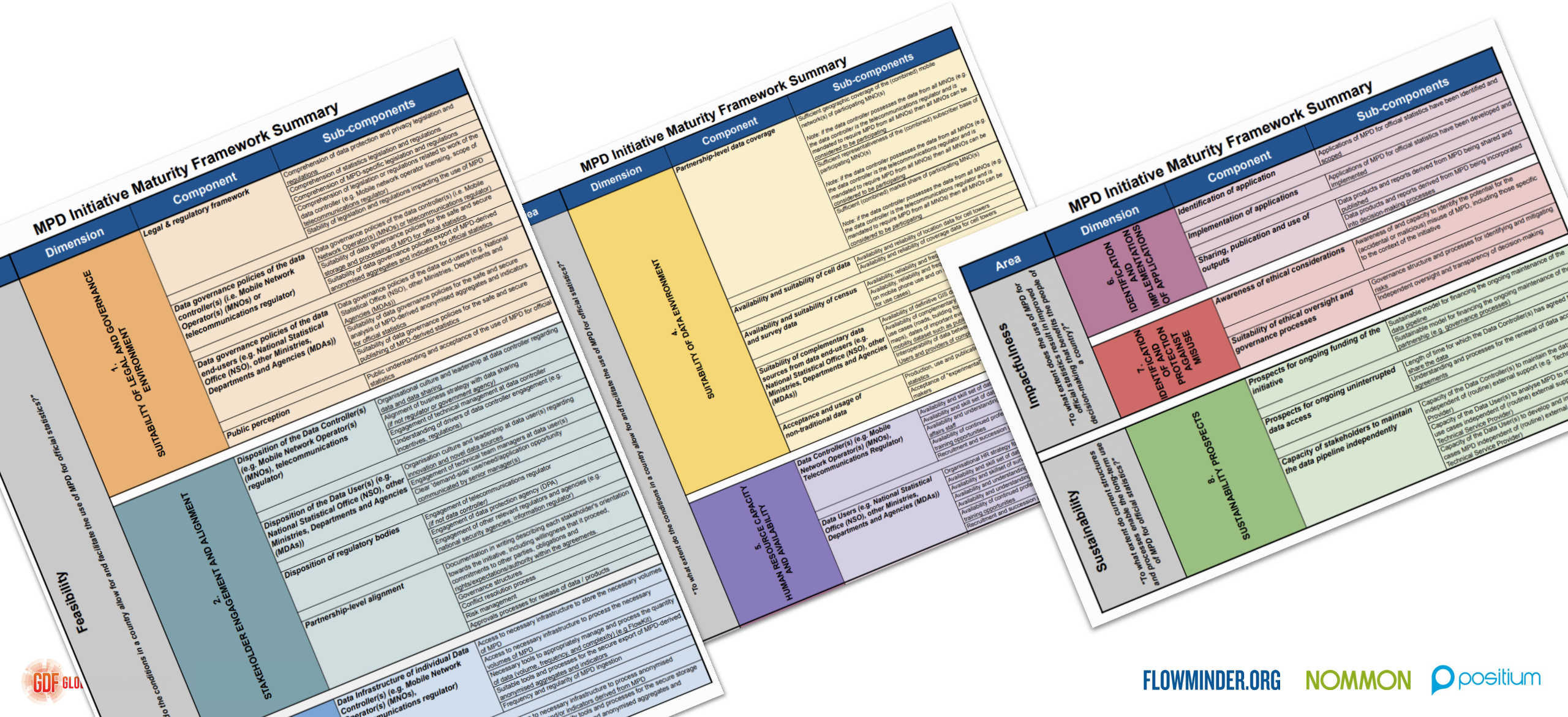


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3 | 8

Maturity Assessment Framework & Tool



Online Training on Foundations in MPD – Data Academy



Foundations in Mobile Phone Data for Policy: Design and Implementation of MPD Initiatives for practitioners Live

<https://courses.wbginstitute.org/course/mobile-phone-data>

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Development Economics

HOME COURSES LEARNING SERIES▼ IMPACT CONT

Foundations in Mobile Phone Data (MPD) for Policy – Design and Implementation of MPD Initiatives



er Training Course is designed to get
tly on using mobile phone meta-data
g. It demystifies what the data can
en policy applications, and walks
tical steps needed to get started—
and expertise to develop and



Technical Foundations in Mobile Phone Data for Policy: Data Pipelines and Product

<https://courses.wbginstitute.org/course/clone-foundations-in-mobile-phone-data-for-policy-design-and-imp>

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HOME COUF

Technical Foundations in Mobile Phone Data (MPD) for Policy – Data Pipelines and Data Products

This training program is designed for technical stakeholders involved in creating or expanding the use of mobile phone data for Policy in their country context. Participants are expected to include statisticians, data analysts, data engineers, data scientists and technical project managers. The course will deliver hands-on knowledge on MPD pipelines, data processing, and data products.



GDF-MPD Knowledge Portal:



Welcome to the GDF-MPD Documentation Portal

Essential Resources

Theory of Change, Maturity Assessment, MoU templates, and more.

Methods & Code

Methodology description and code for specific use cases such as Tourism statistics, Migration statistics, Urban mobility.

Workshops

Resources shared during in-person workshops

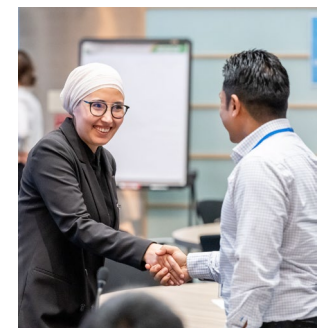
Training material

Key presentations from Foundations Training courses

<https://worldbank.github.io/GDF-MPD/homepage.html>

Work Together

There will always be aspects that differ depending on the local context – The technical aspects are not the hard part





Partner With Us!



Email

gdfmpd@worldbank.org



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GOBIERNO
DE ESPAÑA

MINISTERIO
DE ECONOMÍA, COMERCIO
Y EMPRESA



NATIONAL
STATISTICS OFFICE
OF MONGOLIA



MPD - MONGOLIA:

Operationalization for official statistics

NEW YORK, USA
4 MARCH 2026





BRIEF INTRODUCTION OF THE PROJECT: Country experiences

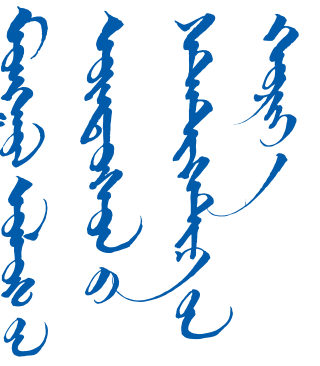


The World Bank's Global Data Facility –Mobile Phone Data (GDF–MPD) Program

Objectives:

- ✓ Strengthen national capacities for MPD use and governance;
- ✓ Accelerate sustainable operational processes.





BRIEF INTRODUCTION OF THE PROJECT: Country experiences

Global MPD Usage

PUBLIC SERVICE

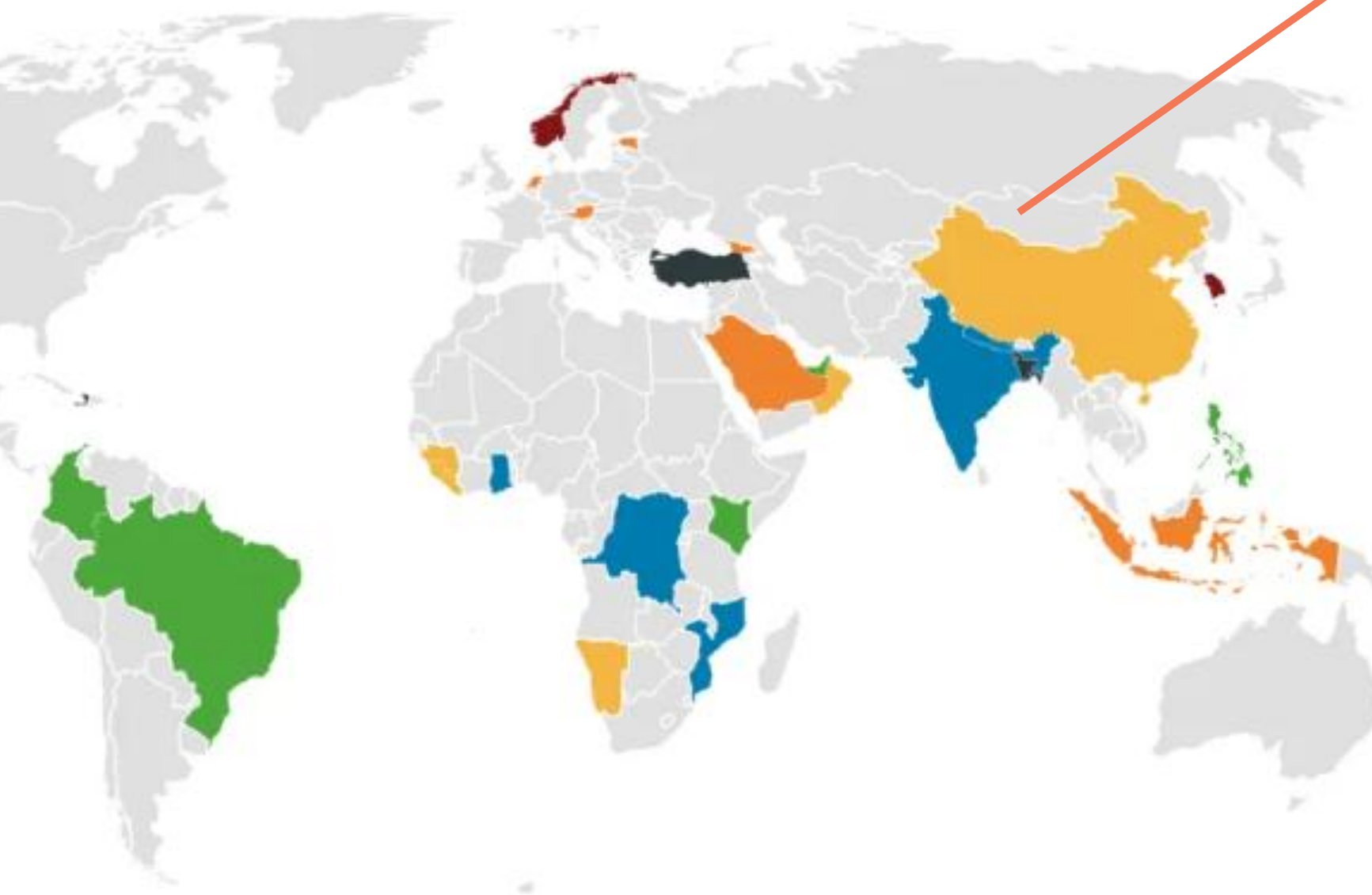
By determining the location of the population **Improving access to schools, kindergartens, hospitals, and other public service institutions**

HEALTH SECTOR

Disease prevalence, **Real-time monitoring of dangerous health conditions** and implement responses quickly and effectively

URBAN PLANNING AND DEVELOPMENT

Public transportation, housing, new roads, facilities, etc. **Identify needs of infrastructure**



1. INTERNAL POPULATION MOBILITY

Analyze patterns of population mobility, monitor their changes, and make decisions accordingly

2. TOURISM INDUSTRY

Study of travel **region**, and travel flow and to **plan and manage work and services of tourism organization more effectively**

INFORMATION AND COMMUNICATION STATISTICS

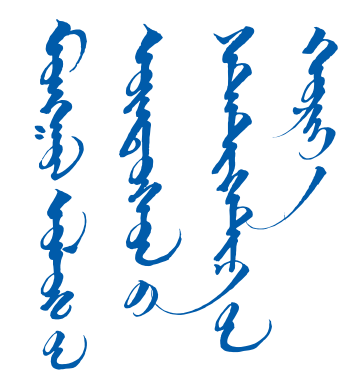
Create baseline information to inform the formulation of policies and programs at national and international levels, and to assess their implementation.

TRANSPORT INDUSTRY

Planning of public a transportation route and monitoring of traffic flow, congestion, and safety.

DISASTER RESPONSE MEASURES

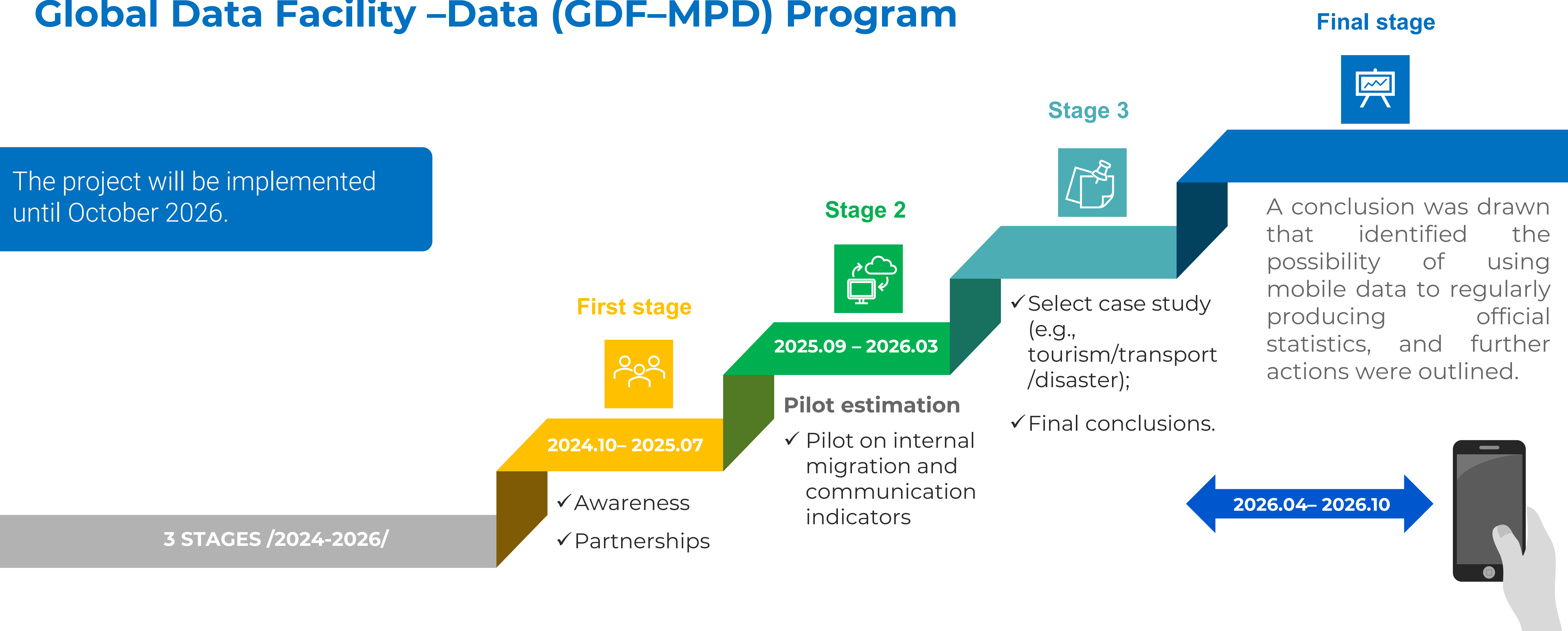
Monitor population movement during **emergencies** to identify areas in need of **assistance and respond to disasters.**



BRIEF INTRODUCTION OF THE PROJECT: Country experiences

Global Data Facility –Data (GDF–MPD) Program

The project will be implemented until October 2026.





PARTNERSHIP MODELS BETWEEN NSOs AND OTHER INSTITUTIONS



The World Bank provides technical assistance.



Positium provides the methodological support required for the implementing the project, including data processing infrastructure and mobility analysis methodologies.



WORKING GROUP

A working group was established by the joint order of 2024 issued by the Chairman of the NSO and the Chairman of the Communications Regulatory Commission (CRC).



READINESS ASSESSMENT

An assessment of readiness to leverage MPD was performed by international consultants for mobile network operators (MNO), the CRC, and the NSO on March 13-14, 2025.



CONSULTATION MEETING AND TECHNICAL SEMINAR

A consultative meeting and technical seminar to enhance stakeholders' understanding of MPD usage was organized on March 18-19, 2025, with the participation of 105 representatives from the NSO, the CRC, the National Human Rights Commission, relevant ministries and agencies, and MNO.

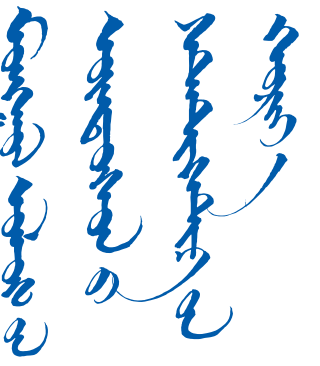


MEMORANDUM

On July 2, 2025, a Memorandum of Understanding on cooperation was signed between the Chairman of the NSO, the CRC, and Mobicom Corporation LLC.



APPROACHES TO BUILDING DURABLE AND SUSTAINABLE MPD PIPELINES



Secure Processing Environment & Data Handling



Full codebase, methodology, and workflows formally transferred to NSO internal ownership for independent, long-term use.



Knowledge & Technology Transfer



Institutional independence achieved for pipeline operation, updates, and scaling without ongoing external support.



Capacity Building & Skills Development



Skilled core team ready to independently run, troubleshoot, and expand MPD pipelines.



Methodological Advancements for Internal Population Migration



Foundation ready for producing official/experimental internal migration statistics.

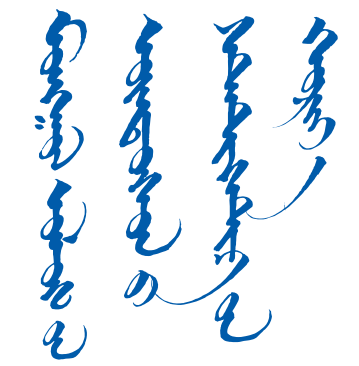


ICT Sector Big Data Applications



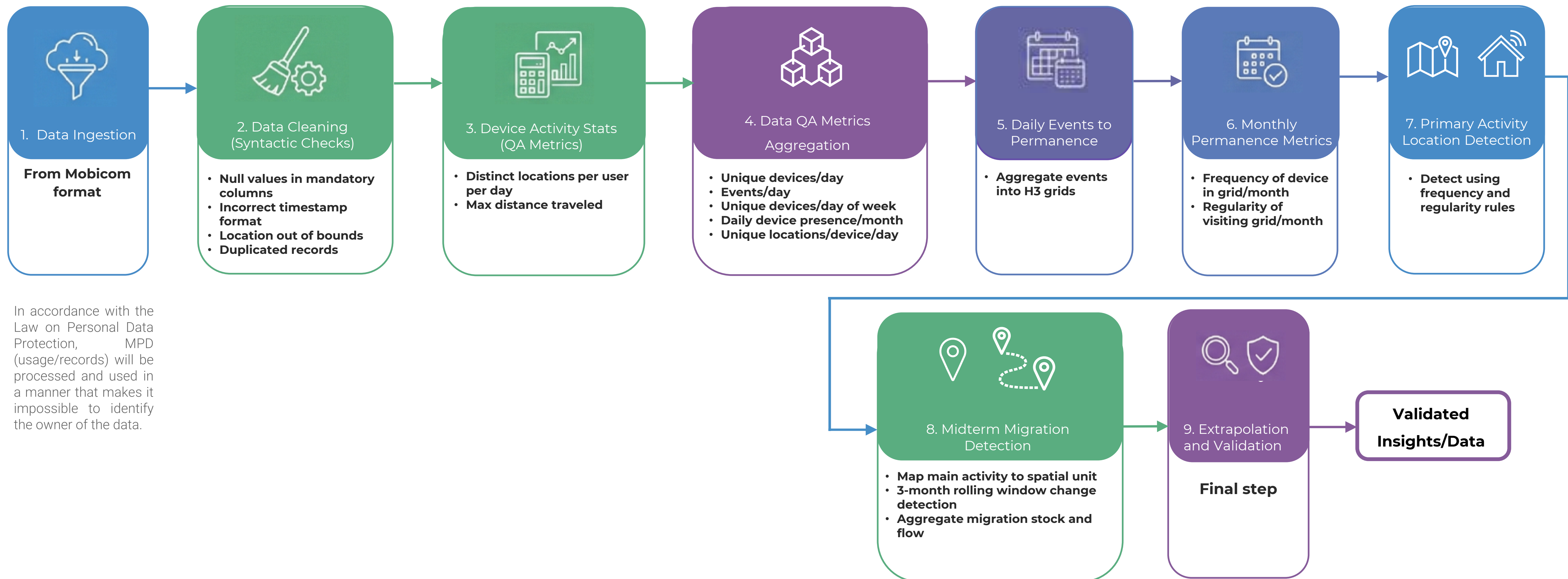
NSO-owned, documented methodology and pipeline for regular, timely production of MPD-derived ICT statistics.



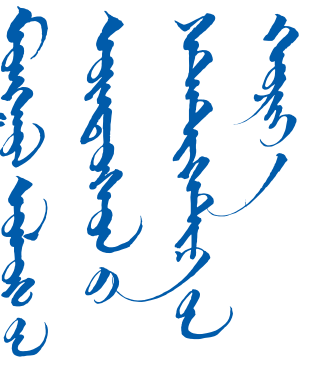


APPROACHES TO BUILDING DURABLE AND SUSTAINABLE MPD PIPELINES

Mobility Data Processing & Migration Detection Workflow



In accordance with the Law on Personal Data Protection, MPD (usage/records) will be processed and used in a manner that makes it impossible to identify the owner of the data.



APPLICATIONS USING MPD FOR OFFICIAL STATISTICS

2ND STAGE OF THE PROJECT

Started from September 2025 to the end of the March

INTERNAL POPULATION MOBILITY



3RD STAGE OF THE PROJECT

2026 April to October

DOMESTIC TOURISM

- Study of travel **region**, and travel **flow**



THANK YOU FOR
YOUR ATTENTION



Using the potential of mobile positioning data

IN OFFICIAL STATISTICS

The experience of Kazakhstan

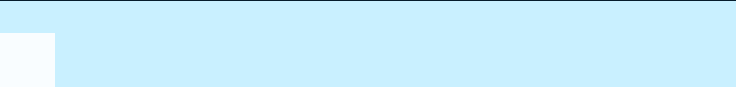
Maxat Turlubayev

2026



STAGES OF MPD IMPLEMENTATION

2023



- Study of international experience and international recommendations on the use of MPD in statistics
- Initiation of amendments to the Law of the Republic of Kazakhstan “On State Statistics” to obtain access to raw data (MPD)



2024



- Amendments to the Law of the Republic of Kazakhstan “On State Statistics” partially approved:
- The Bureau receives MPD based on an agreement with OSS
- Start of cooperation between the Bureau and OSS
- Signing of an agreement with Kcell (market share – 29.9%);
- Receipt of a 5% MPD sample from Kcell
- Preparation of scripts for cleaning and processing MPD
- Compilation of experimental statistics;

2025



- Amendments to the Law of the Republic of Kazakhstan “On Communications” regarding obtaining MPD for statistical purposes
- Development and approval of rules for the provision of anonymized subscriber data for statistical purposes to the Bureau
- Development of technical specifications for creating an information system for data processing
- Creation of a secure data processing model in a “sandbox” environment using data samples

IMPORTANT LEGISLATIVE CHANGES IN THE REPUBLIC OF KAZAKHSTAN

01 Active use of administrative and alternative data sources

According to Article 12 of the Law of the Republic of Kazakhstan “On State Statistics”, **BNS uses administrative and alternative data for the production of official statistical information** and for updating statistical registers.

According to Article 24 of this Law, **alternative data are provided** by their owners or holders in accordance with agreements on the provision of alternative data to the **authorized body for state statistics on a free-of-charge basis**.

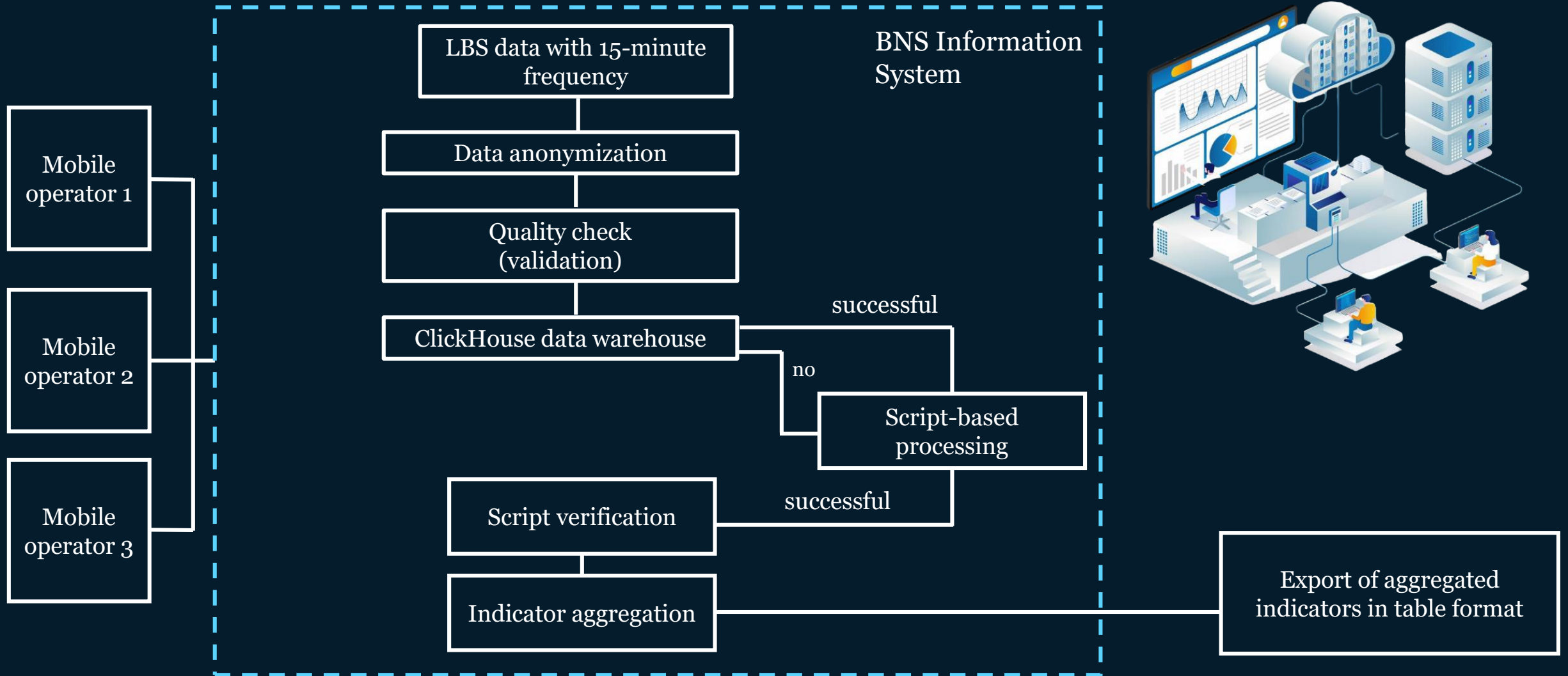
02 Obtaining anonymized data from mobile subscribers for statistical purposes

According to Article 15-3 of the Law of the Republic of Kazakhstan “On Communications”, **telecom operators and network owners provide anonymized subscriber data to the authorized body for state statistics**.

According to the “**Rules for the Provision by Telecom Operators and (or) Network Owners of Anonymized Subscriber Data for the Implementation of State Statistical Activities**”, a unified procedure, requirements, and conditions for the provision of anonymized data have been established, including measures to ensure confidentiality and personal data protection.



INTERACTION MODEL WITH OPERATORS



USE OF MOBILE POSITIONING DATA FOR TOURISM STATISTICS



BASED ON MOBILE POSITIONING DATA



Respondent burden:
8 thousand accommodation facilities
21 thousand households
10 thousand foreigners

Publication **2 months** after
data collection

Does not cover
unregistered flows

Limited
information



BASED ON STATISTICAL FORM DATA

Reduction of
respondent burden:
data from **3 mobile**
operators

Publication
approximately **2 weeks**
after data collection

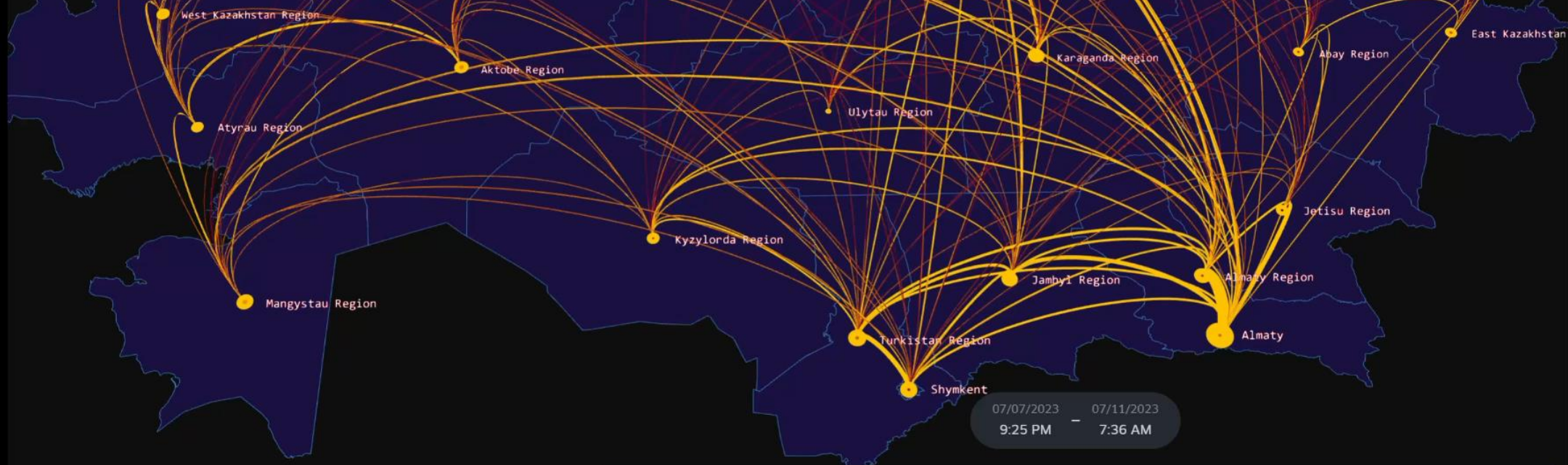
Coverage ~ **100% of the**
population (including
“invisible” tourists)

AI-enriched data:
behavioral patterns and
commuting migration



The video shows domestic tourism of Kazakhstan based on mobile positioning data

July 2023 – June 2024



The Republic of Kazakhstan

The visitors over the period:

-official statistics – 7.8 mln people

-mobile data – 107.6 mln people (14 times)



ADVANTAGES OF USING MPD FOR STATISTICAL PURPOSES

1

High data frequency, geographic granularity, broad population coverage, inclusion of all visitors, including self-organized tourists.

2

Consistency with official statistics in structure and trends, demonstrating the reliability and credibility of mobile positioning data obtained from telecom operators (OSS).

3

Ability to complement census and population register results, enabling a shift from static to dynamic analysis of settlement systems.

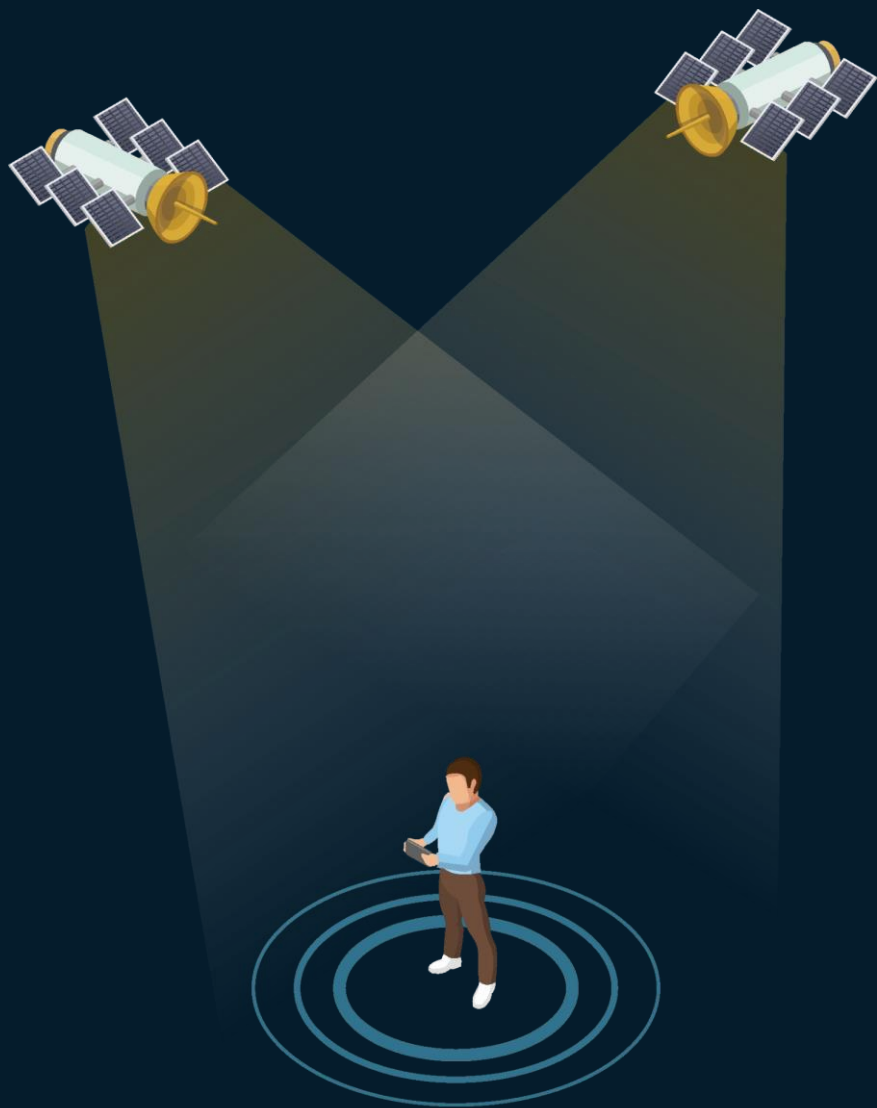
4

A valuable source of information on the age and gender structure of visitors, their portrait, geography of stay, and preferences in choosing tourist destinations.

5

An effective and promising tool for analyzing and managing tourist flows and for the development of the tourism industry.





NEXT STEPS



Receiving 100% of data from all 3 mobile operators



Development and approval of the methodology for compiling official statistics indicators based on MPD



Integration of MPD for developing new areas of official statistics (migration, transport, information society)



Reducing respondent burden, optimizing statistical surveys and population censuses



Improving timeliness, granularity, and reliability of statistical indicators

CONCLUSIONS



- 1** MPD is not a replacement but a complement to traditional sources of official statistics.
- 2** Effective use of MPD requires a legal framework, technological readiness, and cooperation with the private sector.
- 3** International cooperation and exchange of experience will help accelerate the implementation of innovative approaches.
- 4** Pilot projects are an important tool for testing methodologies and assessing data quality.
- 5** Development of digital skills and strengthening institutional capacity are key to the successful transformation of statistical systems.

**THANK YOU FOR
YOUR ATTENTION**



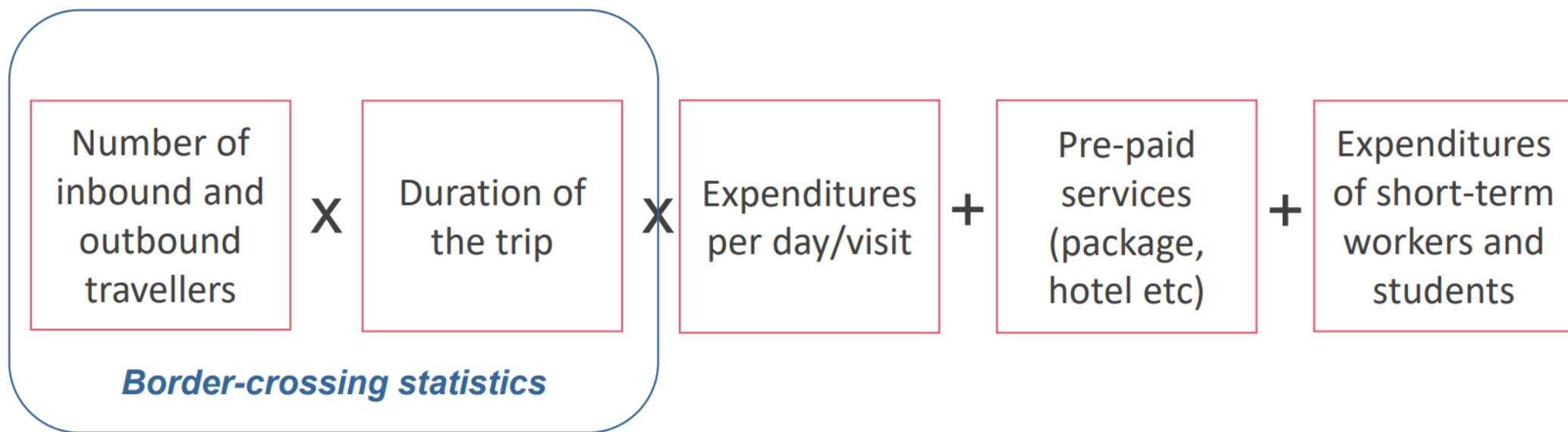
MPD for Statistics

the case of Estonia



MPD for Tourism Statistics, Bank of Estonia

- Cross-border travel statistics – part of balance of payments
- Fully anonymised data
- Calibrated with official sources
- Long-term cooperation with mobile operators



Mobile Phone Data: Advantages



Timeliness (in Estonia T+18)



Consistency



Completeness/coverage



Cost-efficiency

Mobile Phone Data: Challenges



No universal standardised methodological foundations



Models need to be re-estimated according to the changes in economic environment and technological adoption



Need to be up-to-date on privacy laws (GDPR vs old telecom law) and ensure continuous access to the big-data

Where to next?



More granular mobility insights



Combining telecom data with statistical registers



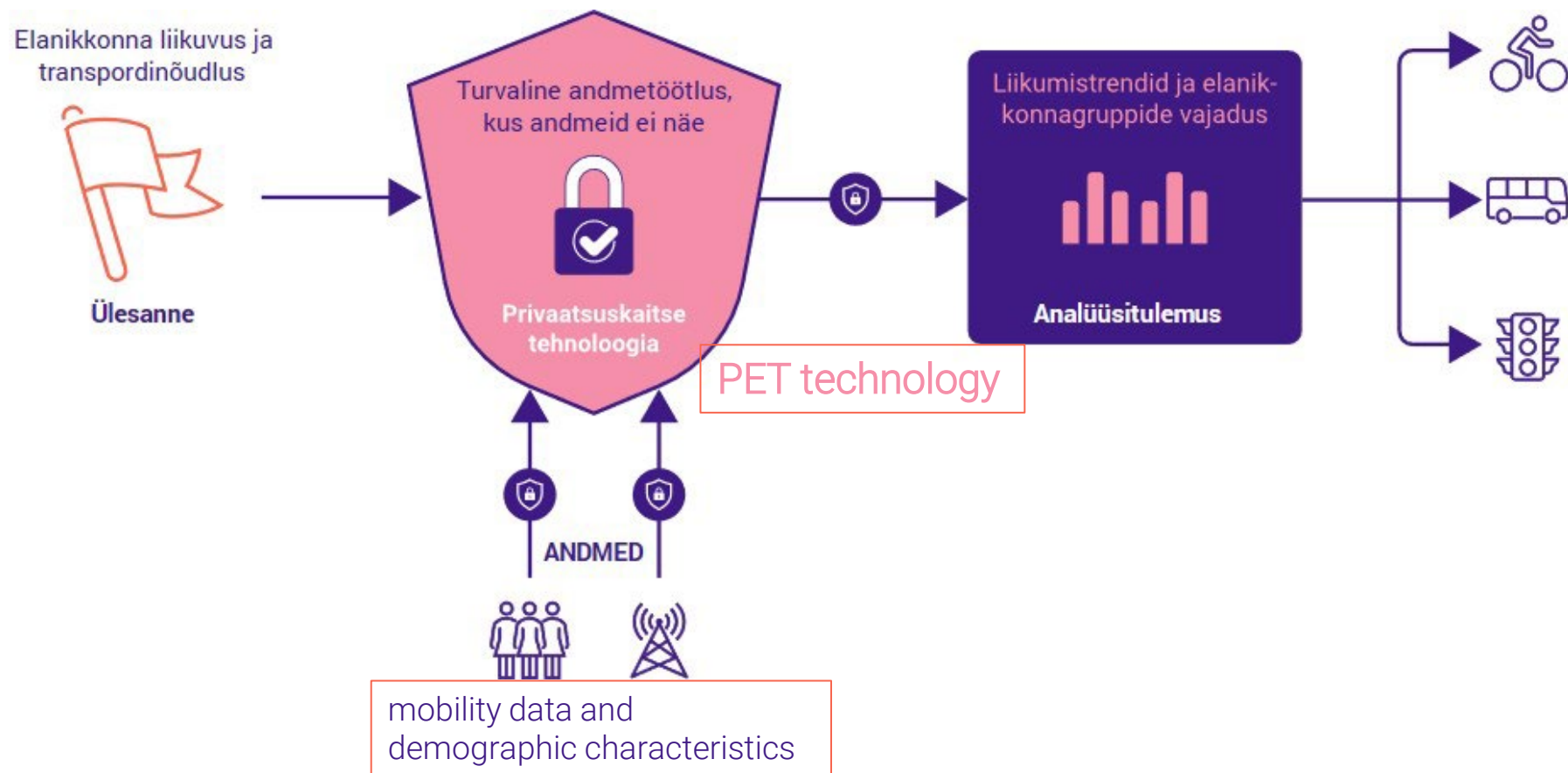
Two strict legal regimes. How to satisfy both simultaneously?

Use Case 2: Secure Cross-Use of Data between Sectors

population mobility and
transport demand

processing of different parties' data
without any disclosure of personal data

mobility trends and needs of different
demographic groups



From Data Sharing to Algorithm Sharing



Single joint anonymisation pipeline



Privacy-Enhancing Technologies (PETs)



Secure, jointly governed computation

Example: Tallinn–Tartu Mobility Corridor

- Which user profiles use 2+2 highway vs rail?
- Are low-income groups dependent on certain routes?
- How does mobility differ by age structure, household type, employment status?
- Who benefits most from infrastructure upgrades?

Infrastructure planning becomes socially informed, not just traffic-volume driven.



Beyond One Project. A New Governance Model for Official Statistics



Scalable public–private data reuse model



Trust-based partnerships



Technical enforceability



Embedded privacy protection



Contributing to global dialogue

Key takeaways from our experience



Partnership must go beyond data transfer



Public communication matters



Capacity constraints are real

Thank you!